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Master's Thesis

Decoding Spatial Pain Anticipation from EEG in a Pavlovian Threat-Learning Paradigm

Research carried out in collaboration with the Department of Psychology, Center for Studies and Research in Cognitive Neuroscience

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Abstract

Pain anticipation is not merely a general expectation that something unpleasant may occur. In many situations, the expected bodily location matters as well, because preparation can then be adjusted to the anticipated site of harm. This thesis asks whether EEG recorded during the cue period of a Pavlovian threat-learning task contains enough information to decode that spatial content of anticipation, distinguishing left-threat, right-threat, and safety at the single-epoch level.

The study focuses on Block 1, the acquisition phase in which cue meaning is most stable. Starting from raw annotated BrainVision recordings, it reconstructs ordered boundary-to-boundary segments while preserving subject and recording provenance, and then excludes cue epochs invalidated by duration anomalies or DC-related contamination. With a cue-duration tolerance of 0.300 s and a post-DC exclusion tail of 1.836 s, 30 raw Block 1 files yielded 1711 clean cue epochs from 29 exported subjects/recordings, with class counts of 571 left-threat, 572 right-threat, and 568 safety epochs.

Three model families were compared under a grouped subject-wise train/validation/test split with train-only normalization: a baseline temporal convolutional network (TCN), an attention-augmented TCN, and a compact convolutional neural network operating on short-time Fourier transform (STFT) spectrograms. The strongest full-band held-out result was obtained by the attention TCN, which reached 39.93% test accuracy and macro-F1 0.3867, above the balanced three-class chance level of 33.33%. The alpha-restricted attention TCN achieved the strongest held-out result overall, with 40.27% test accuracy and macro-F1 0.3975; theta-restricted decoding remained informative but weaker, and C3/C4-only decoding was weaker overall.

Taken together, the results support a cautious but positive answer to the thesis question: under leakage-free subject-wise evaluation, cleaned cue-period EEG contains modest yet meaningful information about the expected spatial content of pain. The thesis therefore contributes both an empirical decoding result and a transparent methodological framework for spatially specific EEG decoding in artifact-prone multimodal recordings.

List of Abbreviations

AI	Artificial Intelligence
CS	Conditioned stimulus
CS+L	Cue associated with expected pain on the left arm
CS+R	Cue associated with expected pain on the right arm
CS−	Safe cue, never paired with pain
DC	Direct-current correction event annotated in the recording stream
EEG	Electroencephalography
EMG	Electromyography
FDI	First dorsal interosseous muscle
MEP	Motor-evoked potential
MNE	Open-source Python environment for EEG/MEG analysis
NPZ	NumPy compressed array archive
SCR	Skin conductance response
STFT	Short-time Fourier transform
TCN	Temporal convolutional network
TMS	Transcranial magnetic stimulation
US	Unconditioned stimulus (painful electrical stimulation)

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Chapter 1

Introduction

1.1 Motivation

Pain anticipation is a particularly revealing case of prospective brain function. The nervous system does not merely register noxious input after it occurs; it uses expectations, prior learning, and contextual information to shape how potentially harmful events are prepared for, interpreted, and acted upon [18, 27, 19, 6, 20]. In pain, this anticipatory dimension is especially important because the organism benefits from preparing before tissue threat is confirmed. Expectations can alter subjective pain, bias attention, change defensive behaviour, and recruit physiological responses even in advance of the painful event itself [18, 27, 19, 10]. From that perspective, anticipatory activity is not a peripheral prelude to pain processing. It is part of how pain-related meaning is constructed.

The present thesis focuses on a more specific and more demanding version of that problem: the spatial content of pain anticipation. In everyday life, it matters not only whether something harmful is expected, but also where it is expected on the body. Anticipating pain on the left arm is not behaviourally equivalent to anticipating pain on the right arm, because bodily preparation, selective attention, and motor readiness can in principle be tuned to the predicted site of harm [8, 9, 29, 30, 6, 37]. A decoding problem defined by left-threat, right-threat, and safety is therefore scientifically richer than a simple threat-versus-safety contrast. It asks whether anticipatory neural activity contains body-specific information rather than only a coarse signature of aversive expectation.

This question is also methodologically meaningful. The data analysed here were acquired in a multimodal TMS–EEG experiment that additionally measured corticospinal excitability through TMS-induced motor-evoked potentials (MEPs), autonomic responding through skin conductance responses (SCR), and subjective ratings of learning and expectancy [40, 41]. Such a design is scientifically powerful because it situates EEG within a broader account of anticipatory pain processing. At the same time, it makes analytical discipline indispensable. Concurrent TMS–EEG

is sensitive to transient artifacts, decay components, and preprocessing choices, while EEG machine learning is vulnerable to subject leakage and overoptimistic evaluation when train/test separation is not handled conservatively [1, 2, 25, 3, 15, 17, 26]. In this setting, a modest above-chance result obtained under strong controls can be more informative than a larger score obtained under weaker ones.

The broader project context makes the thesis question even more compelling. The internal experiment illustration and poster describe a design in which acquisition and reversal are studied jointly, and in which somatomotor alpha, frontocentral theta, MEPs, and SCR are interpreted as partially complementary anticipatory signatures [40, 41]. At the project level, alpha suppression is described as contralateral to the expected side of shock, theta as more closely related to threat value and reversal-related updating, MEP inhibition as limb-specific, and SCR as a non-lateralized autonomic index of threat [4, 5, 40, 41]. The present thesis does not reanalyse every modality. Instead, it asks whether cue-period EEG alone already contains enough information to recover the expected spatial content of impending pain at the single-epoch level. That question sits naturally at the intersection of neuroscience and machine-learning methodology.

1.2 Problem statement

The central question of the thesis is the following:

> Can EEG activity recorded during the cue period in Block 1 of a Pavlovian threat-learning experiment be used to decode the expected spatial content of pain, distinguishing left-threat, right-threat, and safety at the single-epoch level under a leakage-free subject-wise evaluation protocol?

This is narrower than a general "can EEG classify the task?" formulation, and that narrowing is deliberate. First, the interval of interest is restricted to the anticipatory cue period rather than to post-stimulus outcome processing. Second, the analysis is restricted to Block 1 rather than to the full acquisition-plus-reversal experiment. Third, the evaluation strategy is intentionally conservative: splitting is grouped by subject, normalization is estimated from training data only, and final claims are anchored to held-out performance rather than to validation behaviour alone [15, 16, 17, 26]. Fourth, the thesis compares matched time-domain and time–frequency model families on the same cleaned source dataset rather than presenting a single architecture in isolation.

That framing matters scientifically. A threat-versus-safety classifier could perform above chance while remaining agnostic about bodily location. The present question is more stringent because it asks whether the anticipatory state is spatially structured enough to distinguish left-threat from right-threat while still separating both from safety. It also matters methodologically because the target is subtle, the dataset is modest, and the recording context is artifact-prone. Under those conditions, the goal is not simply to maximize a headline metric, but to determine whether a carefully constructed and leakage-controlled analysis yields evidence that survives held-out subject-wise evaluation.

In practical terms, the problem decomposes into five connected tasks:

1. reconstructing a segment-level dataset from raw annotations while preserving subject, recording, and temporal provenance;
2. identifying and removing cue epochs invalidated by DC-related contamination or by clear duration anomalies;
3. exporting fixed-size time-domain and STFT representations from the same retained epochs;
4. enforcing a split and normalization protocol that avoids optimistic bias due to subject leakage or test-set peeking; and
5. interpreting the final performance in light of both the neuroscientific question and the methodological limits of the dataset.

1.3 Why the thesis remains focused on Block 1

The broader experimental design includes both acquisition and reversal. Reversal is theoretically important because it probes updating, uncertainty, competition between old and new cue contingencies, and the reorganisation of threat predictions over time [23, 24, 28, 4, 5]. Precisely for that reason, however, it is not the simplest setting in which to establish a first tightly controlled decoding analysis. Once contingencies reverse, neural activity may reflect a mixture of current expectancy, residual learning from the original mapping, surprise, control-related updating, and uncertainty about the new associations [23, 24]. Those features are scientifically valuable, but they make interpretation of the target labels less direct.

Block 1 is the phase in which cue meaning is most stable. One cue predicts left-sided stimulation, one predicts right-sided stimulation, and one predicts safety. The target labels are therefore aligned more directly with the anticipatory state of

interest. This choice is not meant to diminish the importance of reversal. Rather, it reflects a methodological sequencing principle: before moving to the more complex problem of changing contingencies, it is useful to establish whether the spatial content of anticipation is decodable in the phase where cue meaning is most stable and least confounded by relearning dynamics [40, 41]. Focusing on Block 1 therefore strengthens interpretability while also providing a rigorous test bed for the cleaning and evaluation strategy used throughout the thesis.

1.4 Broader significance of the work

The significance of the thesis is both neuroscientific and methodological. From a neuroscience perspective, successful decoding of left-threat, right-threat, and safety would suggest that cue-period EEG carries information not only about the presence of threat, but also about its predicted bodily location. That would support the idea that anticipatory pain processing is spatially structured and at least partly embodied, rather than limited to a generic arousal or aversiveness signal [6, 29, 30, 8, 9, 4]. It would also resonate with the broader project picture in which somatomotor alpha and corticospinal excitability both track the anticipated side of harm, whereas SCR reflects a broader threat-versus-safety contrast [40, 41, 34, 33].

From a methodological perspective, the thesis addresses a common tension in EEG machine learning. Flexible models can detect subtle structure in complex signals, but encouraging performance can easily become difficult to interpret when preprocessing is opaque, subject leakage is present, or artifact handling is underjustified [15, 16, 17, 26]. The present study therefore treats data construction, contamination handling, provenance, and evaluation design as part of the scientific contribution rather than as invisible preprocessing. Under such conditions, even modest above-chance performance can still be meaningful, because it is obtained under constraints designed to protect interpretability.

1.5 Main contributions

This thesis makes four interrelated contributions.

First, it builds a traceable preprocessing and export pipeline for Block 1 cue-period epochs. Every retained sample carries subject identity, block identity, recording identity, within-recording order, annotation context, and source-file provenance.

Second, it treats DC contamination as an explicit object of analysis. Rather

than assuming that the annotation boundary of a DC-correction event fully captures its effect on the signal, the study estimates a post-DC exclusion tail from dedicated signal-based recovery analyses and then applies that threshold in the state-machine cleaning logic.

Third, it evaluates models under a grouped subject-wise protocol with train-only normalization, validation-based model selection, and a held-out test set with zero subject and recording overlap.

Fourth, it compares multiple representational and architectural choices without changing the cleaned source dataset: a baseline TCN, an attention-augmented TCN, and a spectrogram CNN in the full-band run, together with alpha-only, theta-only, and C3/C4-only exploratory variants.

1.6 Structure of the thesis

The remainder of the dissertation is organized as follows. Chapter 2 reviews the neuroscientific and methodological background and situates the thesis in the relevant literature on pain prediction, threat learning, oscillatory dynamics, TMS–EEG methodology, and EEG decoding. Chapter 3 describes the broader experiment, the analytical scope retained for the present study, and the role of the auxiliary analyses that support the final dataset and thresholds. Chapter 4 presents segmentation, cleaning, threshold justification, signal extraction, and dataset construction. Chapter 5 explains the representations, split strategy, normalization protocol, model families, training procedure, hyperparameter search, and evaluation metrics. Chapter 6 reports the dataset statistics and decoding results. Chapter 7 discusses the implications of the findings and the main methodological limitations. Chapter 8 concludes. The appendices provide reproducibility notes, search grids, selected configurations, and supplementary diagnostics.

Chapter 2

Theoretical Background and Related Work

2.1 Pain anticipation as predictive and embodied processing

Pain is increasingly understood as a process shaped by expectation, context, valuation, attention, and action relevance rather than as a passive readout of tissue status alone [18, 27, 19, 6, 20]. Expectations can influence pain before sensory confirmation is available, and anticipatory states can alter both subjective experience and physiological readiness [18, 27, 19, 10]. In that sense, anticipation is not simply a prelude to pain processing. It is part of the way the nervous system organises potentially harmful events.

Predictive views of pain are especially relevant here because the present thesis studies cue-period activity rather than the sensory response to delivered stimulation. The cue period is precisely the interval in which the organism prepares for a harmful event on the basis of prior learning. If pain is construed as a process that integrates prediction with incoming input, then cue-related activity is scientifically meaningful in its own right rather than merely preparatory noise before the actual event [18, 27, 20]. This logic also helps explain why even moderate subject-wise decoding performance can matter: the target is a cognitive and bodily state, not a large evoked response to a fully delivered stimulus.

The current thesis adopts an explicitly embodied version of that predictive perspective. The goal is not simply to distinguish threat from safety, but to test whether cue-period EEG contains enough information to differentiate the expected spatial content of a future painful event. This emphasis is important because bodily location is behaviourally consequential: preparing for pain on the left arm is not equivalent to preparing for pain on the right arm. If anticipation is spatially structured, then some of that structure should in principle be detectable in neural dynamics related to sensorimotor preparation and selective bodily attention [8, 9, 6].

2.2 Spatially specific threat representation and bodily anticipation

A growing body of work supports the idea that threat anticipation can be spatially anchored to the body. In experimental pain paradigms, anticipating pain at a particular body location can bias tactile processing toward that threatened site, consistent with the idea that attention and readiness are organized around the expected locus of harm [29, 30]. Importantly, this spatial coding is not necessarily confined to a mathematical point on the skin. Vanden Bulcke and colleagues showed that anticipatory prioritization can extend across a body part and even generalize to different locations on the same side of the body, suggesting that bodily threat is represented at a functional rather than purely punctual spatial scale [30].

This literature is highly relevant for the present thesis. It provides a behavioural and cognitive rationale for using left-threat and right-threat as distinct classes. If anticipation of left-sided pain biases processing differently from anticipation of right-sided pain, then a decoding problem that respects that distinction is better aligned with the structure of the underlying phenomenon than a simpler binary threat-versus-safety task. The question is therefore not merely whether the brain distinguishes threat from safety, but whether the anticipatory state contains enough spatial organization to support differentiation between threat to one side of the body and threat to the other.

The broader project materials point in the same direction. The experiment illustration and poster frame the study not just as fear conditioning in general, but as the construction of pain predictions in the motor system, with explicit attention to where on the body pain is expected [40, 41]. That framing is central to the thesis: the spatial target is not an arbitrary relabelling of the data, but a theoretically motivated operationalization of embodied threat prediction.

2.3 Pavlovian threat learning, acquisition, and reversal

Pavlovian threat learning provides a natural framework for studying anticipatory states. Neutral cues acquire predictive value through repeated association with an aversive outcome, and once these associations are learned, conditioned stimuli can evoke neural, behavioural, and physiological responses before the painful event occurs [21, 22, 28]. In the present project context, two coloured cues predicted pain on the

left or right arm, whereas a third cue predicted safety. Reinforcement was partial: each threat-predicting cue was followed by pain on 70% of its trials and omitted on the remaining 30% [40, 41]. This partial-reinforcement structure matters because the target of the present study is the anticipatory state evoked by the cue, not the eventual presence or absence of stimulation on a single trial.

Acquisition and reversal are not interchangeable phases of the same process. Acquisition is the period in which the cue–outcome mapping is established. Reversal introduces a qualitatively different problem, because the organism must inhibit or revise previous associations while learning new ones [23, 24, 28]. Neuroimaging work on conditioning and reversal repeatedly emphasizes that reversal engages expectancy updating, behavioural flexibility, and the negotiation between prior and current contingencies [22, 23, 24]. In the internal project materials associated with this thesis, the same logic appears at the design level: page 3 of the experiment illustration lays out a sequence of preparation, baselines, acquisition, ratings, reversal, and post-learning baseline, while the poster explicitly frames reversal as the phase in which learned threat predictions must be recalibrated [40, 41].

That distinction is crucial for the present thesis. The decoding target is easier to interpret in acquisition because cue meaning is stable. During reversal, the same EEG pattern could reflect current prediction, residual learning from the earlier mapping, uncertainty about the remapped contingencies, or the active updating process itself. Restricting the present decoding analysis to Block 1 is therefore not a convenience choice but a principled way to align the labels as closely as possible with the anticipatory state of interest.

2.4 Alpha-band dynamics, somatomotor preparation, and bodily specificity

Among the candidate EEG signatures relevant to this thesis, somatomotor alpha is particularly important. Alpha-band activity over sensorimotor cortex has been linked to anticipatory regulation, selective bodily preparation, and pain expectation [7, 8, 9]. More generally, anticipatory alpha modulation has long been interpreted as part of the functional gating of sensory and sensorimotor processing, with contralateral decreases and ipsilateral increases reflecting changes in regional engagement and suppression depending on task demands [31]. In pain-related settings, alpha modulation has been associated both with the expectancy of aversive stimulation and with the spatially organized preparation that precedes it [7, 8]. This makes alpha a strong theoretical

candidate for carrying information about whether threat is anticipated on the left or on the right.

The broader project context reinforces that expectation. The internal experiment illustration and poster interpret motor alpha suppression as a central marker of threat anticipation, with stronger suppression for threat than for safety and stronger contralateral modulation relative to the expected side of pain in both acquisition and reversal [40, 41]. The published project papers likewise describe alpha as tracking both threat value and its lateralized bodily content [4, 5]. This provides a direct theoretical bridge to the present thesis: if alpha carries body-specific anticipatory information at the project level, then alpha-focused decoding analyses should be especially informative.

The motor interpretation is strengthened further by the relation to corticospinal excitability. Both the experiment illustration and the poster report that MEPs are reduced for threat cues selectively on the limb where pain is expected and that alpha, but not theta, predicts MEP changes [40, 41]. Related work outside the project has likewise shown that pain expectancy can modulate corticospinal excitability and autonomic responding in advance of actual stimulation [33], and the recent project-linked MEP study further supports the idea that pain-related motor adaptations can be topographically selective and anticipatory [34]. Taken together, these converging findings make alpha a plausible carrier of spatially specific anticipatory information rather than a merely generic arousal marker.

2.5 Theta-band dynamics, threat processing, control, and updating

Frontocentral theta is also theoretically relevant, but probably for somewhat different reasons. Midfrontal or frontocentral theta has often been linked to conflict monitoring, uncertainty, control allocation, and adaptive updating [32]. In the context of threat learning, these functions are especially relevant when cue meaning is being acquired, challenged, or revised. Theta may therefore be less tightly tied to bodily laterality than alpha and more strongly linked to the broader management of prediction, control, and behavioural significance [4, 5, 23, 24].

This interpretation is consistent with the internal project materials. The experiment illustration describes frontocentral theta as increasing for threat cues relative to safety during acquisition regardless of expected pain side, and as showing a broader increase from acquisition to reversal [40]. The poster makes the same point in more

condensed form, emphasizing that theta reflects threat during acquisition and the updating of pain predictions during reversal [41]. The published project papers likewise link theta to acquisition-versus-reversal differences, cingulate involvement, and more global threat-related dynamics rather than to a narrowly lateralized motor code [4, 5].

For the present thesis, theta therefore remains relevant even if it is not the strongest candidate for body-specific decoding. It may still support classification by helping distinguish threatening from safe anticipatory states, by carrying information about control-related engagement, or by reflecting expectancy-related processes that generalize across cue types. Comparing alpha-restricted and theta-restricted analyses is therefore informative not only for performance ranking but also for theory: it helps distinguish between a predominantly bodily preparatory account and a more general threat-related control account.

2.6 Multimodal project findings and their relevance for the present thesis

Although the decoding analysis reported here is restricted to EEG, the broader project is explicitly multimodal. The experiment illustration and poster describe a design in which EEG, double-coil TMS, MEPs, SCR, and subjective ratings are collected within the same threat-learning framework [40, 41]. This is not incidental background information. It shapes how the EEG findings should be interpreted.

At the project level, the modalities are not redundant. MEPs are described as encoding both threat value and threatened body side, with limb-specific inhibitory modulation congruent with the expected shock location [40, 41]. SCR, by contrast, is described as increasing for threat cues regardless of where pain is expected and regardless of the side from which SCR is recorded, making it a non-lateralized index of autonomic threat responding [40, 41]. The materials also highlight an asymmetric relation between oscillations and motor output: alpha predicts MEP modulation, whereas theta does not [40]. This pattern suggests a functional division in which somatomotor alpha and corticospinal excitability are more directly tied to embodied spatial preparation, while theta and SCR index broader threat-related processes.

That multimodal contrast is valuable for the present thesis because it clarifies what would make the EEG decoding results theoretically interesting. If the best decoding results are obtained in settings that preserve lateralized alpha-related information, that pattern would be consistent with the broader project account. If

performance were dominated instead by non-lateralized signatures, the interpretation would shift toward a more generic threat-versus-safety account. The multimodal project findings therefore provide an external interpretive scaffold against which the EEG-only decoding results can be judged.

2.7 Concurrent TMS–EEG and the methodological cost of contamination

The broader experiment combines EEG with double-coil TMS and MEP measurement. Scientifically, this is a major strength because it allows anticipatory oscillatory activity to be interpreted alongside indices of corticospinal excitability and autonomic responding [40, 41]. Methodologically, however, concurrent TMS–EEG is demanding. TMS introduces large transients, decay components, muscle artifacts, electrode-polarization effects, and substantial preprocessing sensitivity [1, 2, 25, 3]. In practice, seemingly small preprocessing choices can materially affect the signal that remains available for analysis.

In the recordings studied here, this general challenge is compounded by explicit DC-correction events in the annotation stream. The key issue is that the signal disturbance associated with a DC event can remain scientifically relevant after the nominal marker boundary. A cue segment may therefore still be present in the event stream while already being unsuitable as a learning sample. If contaminated epochs are retained, a classifier may learn nuisance structure with little to do with the neural representation of anticipation.

That is why contamination handling is central rather than secondary in this thesis. The post-DC exclusion tail, the duration tolerance, and the state-machine logic are not merely implementation details. They are part of the argument that the final decoding scores should be interpreted as reflecting anticipatory EEG rather than residual artifact structure. This point is especially important in a multimodal setting, where scientific richness and preprocessing difficulty are inseparable.

2.8 Pain decoding, EEG machine learning, and transparent evaluation

Machine learning on EEG has grown rapidly, but the literature also makes clear that reported performance can be strongly distorted by data leakage, unstable cross-

validation, insufficient preprocessing transparency, and small-sample uncertainty [15, 16, 17, 26]. Subject-level leakage is especially important because EEG contains strong participant-specific structure. A model may therefore appear to perform well even when it is relying partly on idiosyncratic subject signatures rather than on the phenomenon of interest.

Pain-related decoding studies illustrate both the promise and the interpretive difficulty of this area. Work using EEG and fMRI has shown that pre-stimulus as well as post-stimulus activity can contribute to the prediction of perceived pain intensity [35]. Multivariate EEG analyses have likewise shown that pre-stimulus activity can carry substantial information about whether later stimulation will be perceived as painful or non-painful [36]. Those studies are important because they establish that ongoing anticipatory or preparatory brain activity can be behaviourally informative. At the same time, they target different problems from the one addressed here: subjective pain perception, pain intensity, or threshold-level categorization rather than the spatial content of cue-induced threat anticipation in a multimodal TMS–EEG dataset.

These differences are not minor. The present thesis focuses on a three-class left-threat/right-threat/safety problem, uses a grouped subject-wise evaluation protocol, and treats cleaning transparency as part of the scientific claim. In that sense, the thesis sits at the intersection of pain neuroscience and methodological caution in EEG machine learning [15, 16, 26]. The aim is not simply to maximize a score, but to establish whether above-chance decoding survives a conservative analytical chain from raw data to held-out evaluation. That combination of spatially specific anticipatory decoding with contamination-aware, leakage-controlled evaluation is the main niche of the present work relative to the broader literature.

Chapter 3

Experimental Paradigm, Dataset, and Analytical Scope

3.1 Overview of the broader experiment

The broader experiment combined EEG, double-coil TMS, MEPs, and SCR to study how Pavlovian threat learning shapes the anticipation of limb-specific pain [40, 41]. According to the internal project poster, the sample included 30 participants, 17 of whom were female, with a mean age of 25.66 ± 6.3 years [41]. The study design included preparation, baseline measurements, an acquisition block, ratings, a reversal block, further ratings, and a final baseline. Within each block, participants completed 20 trials for each cue type, for a total of 60 trials per block [40]. This design allowed the broader project to compare stable acquisition-phase threat prediction with later reversal-phase updating while also relating central EEG measures to corticospinal and autonomic indices [4, 5, 40, 41].

The task design is summarized in Table 3.1. Cue duration was 4.500 s, and the inter-trial interval varied between 8.000 s and 10.000 s. The two threat cues were reinforced on 70% of their trials, corresponding to 14 reinforced and 6 non-reinforced presentations per block. The safe cue was never reinforced. The present thesis focuses only on cue-period EEG from Block 1.

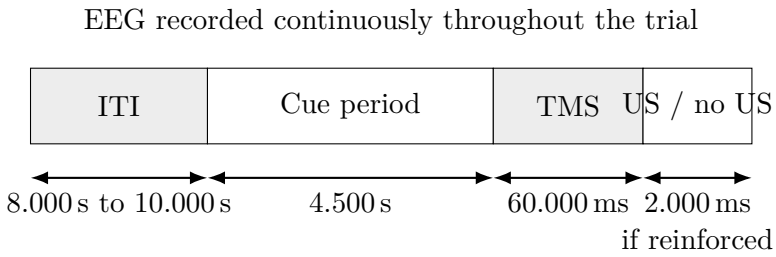

Figure 3.1: Simplified trial structure derived from the internal experiment illustration. The full cue period was treated as the anticipation window of interest in the present thesis.

The internal project materials are useful not only for design description, but also for interpretation. Pages 10–11 of the experiment illustration and the poster

Table 3.1: Summary of the broader experimental design.

Component	Description
Sample	30 participants (17 female; mean age 25.66 ± 6.3 years)
Blocks	Block 1 acquisition; Block 2 reversal
Cue classes	CS+L, CS+R, CS−
Trials per block	20 per cue type; 60 total per block
Threat reinforcement	70% for CS+L and CS+R; 0% for CS−
Cue duration	4.500 s
Inter-trial interval	8.000 s to 10.000 s
Multimodal measures	EEG, TMS-induced MEPs, SCR, subjective ratings
Present thesis	Block 1 cue-period EEG only; three-class decoding of left-threat, right-threat, and safety cues

summarize a limb-specific reduction in MEP amplitude for threat cues, congruent with the expected side of pain, whereas pages 13–14 describe SCR increases for threat cues that are not lateralized by body side [40, 41]. Page 15 of the experiment illustration further highlights that alpha, but not theta, predicts MEP modulation [40]. In other words, the broader project distinguishes between a more body-specific motor signature and a more general autonomic threat signature. This distinction provides an important interpretive backdrop for the present thesis: if EEG decoding is strongest in settings that preserve alpha-related spatial information, that pattern is consistent with the broader multimodal account developed by the project as a whole.

3.2 Operational scope of the study

Although the broader experiment is multimodal and includes reversal, the thesis deliberately focuses on a narrower question. Only Block 1 recordings, task-relevant cue-period segments, and the three-class cue-period decoding problem were carried forward into the main analysis.

Across the Block 1 recordings, 30 raw `.vhdr` files entered the initial scan. From these recordings, 3654 ordered segments were reconstructed. Among them, 1740 were task-candidate cue epochs, meaning segments whose leading annotation corresponded to one of the three target classes before contamination-aware cleaning. After the cleaning rules were applied, 1711 cue epochs remained and 29 were removed. The

resulting machine-learning dataset therefore contained 1711 samples with nearly balanced class counts: 571 left-threat epochs, 572 right-threat epochs, and 568 safety epochs.

The split summary reported 29 unique subjects and 29 unique recording identifiers in the final export, one fewer than the initial raw file count. The evidence therefore supports a precise descriptive statement but not a definitive causal explanation: 30 raw Block 1 files were scanned, whereas the learning dataset contained 29 distinct recording identifiers. One scanned file did not appear as a distinct exported recording in the final learning set, but the evidence does not establish whether this reflects the absence of usable task epochs, an omission during export, or identifier collapse. All split statistics and model results discussed in this thesis are therefore anchored to the dataset that entered model training and evaluation.

Table 3.2: Dataset progression for the final Block 1 analysis.

Stage	Count
Raw Block 1 recordings scanned	30
Ordered segments reconstructed	3654
Task-candidate cue epochs	1740
Removed after cleaning	29
Retained clean task epochs	1711
Exported unique subjects	29
Exported unique recording identifiers	29
Class 0 (left threat)	571
Class 1 (right threat)	572
Class 2 (safety)	568

3.3 Label space and interpretation of the classes

The original stimulus event codes were mapped to the three-class target space shown in Table 3.3. Codes corresponding to real and non-reinforced ("fake" in the implementation) left-arm cues were collapsed into the same left-threat class; the same was done for right-arm cues. This is appropriate because the classifier is intended to decode the anticipatory state elicited by the cue before the outcome occurs. Throughout the remainder of the manuscript, the three classes are referred to as *left-threat*, *right-threat*, and *safety*; code-level names are retained only where exact implementation detail matters.

Table 3.3: Mapping from stimulus event codes to fine labels and final classes.

Event code	Fine label in implementation	Final class
S11	`stim_left_real`	Left threat (class 0)
S110	`stim_left_fake`	Left threat (class 0)
S12	`stim_right_real`	Right threat (class 1)
S120	`stim_right_fake`	Right threat (class 1)
S13	`stim_no`	Safety (class 2)

These code-level labels should not be mistaken for outcome decoding. The model is not asked to predict whether a shock will actually be delivered on a specific trial. Instead, it predicts whether the cue implies expected left-sided pain, expected right-sided pain, or safety during the anticipation period.

3.4 Analytical components and their methodological roles

The thesis draws on a set of complementary analyses with distinct methodological roles. Some were used to estimate contamination-related thresholds, others to characterise the resulting dataset, and others to train and compare the decoding models. Table 3.4 summarizes these complementary roles using concise descriptive labels rather than raw filenames.

Table 3.4: Analytical components and their methodological role in the thesis.

Analytical component	Methodological role	Main contribution to the thesis
Block 1 DC-recovery analysis	Threshold estimation	Provides the block-specific recovery statistics and the earlier 1.827 s recommendation for the post-DC exclusion tail
Combined-block DC-recovery analysis	Threshold estimation	Extends the recovery analysis to the combined-block scan and produces the later shared recommendation of 1.836 s, which then governed the later modelling analyses
Block 1 dataset-audit analysis	Dataset reporting	Summarizes candidate-segment counts, final removals, the stimulus-duration tolerance estimate, and the descriptive audit of the exported Block 1 dataset
Combined-block dataset-audit analysis	Consistency check	Repeats the same style of reporting under a combined-block scan so that the Block 1 report can be interpreted in context
Main Block 1 modelling analysis	Principal analysis	Implements the cleaned export, grouped subject-wise split, full-band time-domain models, full-band spectrogram model, and the principal held-out results reported in the thesis
Alpha-restricted modelling analysis	Exploratory analysis	Repeats the Block 1 analysis after restricting the signal to the alpha band, including both time-domain and spectrogram branches
Theta-restricted modelling analysis	Exploratory analysis	Repeats the Block 1 analysis after restricting the signal to the theta band, again including both time-domain and spectrogram analyses
C3/C4-restricted modelling analysis	Exploratory analysis	Repeats the Block 1 analysis after limiting the input to the two central channels most directly tied to the spatial-motor hypothesis
Auxiliary Block 1+Block 2 modelling analysis	Supplementary analysis	Documents a supplementary Block 1+Block 2 comparison while the main empirical claims remain anchored to the clearly defined Block 1 export

3.5 Configuration choices and analytical scope

One methodological point deserves explicit clarification because it affects how the thresholds are reported. The post-DC exclusion tail was estimated more than once. The Block 1 recovery analysis produced a block-specific recommendation of 1.827 s, whereas the combined-block recovery analysis produced 1.836 s. The classification analyses consistently used the latter value, and the same threshold appears in the alpha-, theta-, and C3/C4-restricted settings. For that reason, 1.836 s is the correct threshold to report for the results discussed in this thesis.

A second point concerns scope. The Block 1 dataset-audit summary draws on an export directory that also contains products from the combined-block analyses, which is why some of its global counts are larger than those of the final modelling dataset. By contrast, the arrays used in the principal modelling analyses contain only Block 1 samples, as shown by the block labels and grouped split summaries. The discussion below therefore distinguishes between descriptive counts from the wider export directory and the specific arrays that entered model training and evaluation.

Table 3.5: Provenance of the main configuration values reported in the present study.

Source	Scope	Key value or scope	Relevance for the thesis
Block 1 DC-recovery analysis	Block 1 only	Recommended post-DC tail = 1.827 s	Earlier block-specific estimate
Combined-block DC-recovery analysis	Combined-block scan	Recommended post-DC tail = 1.836 s	Later shared recommendation written to the configuration path used by the modelling analyses
Main Block 1 modelling analysis	Confirmatory analysis	Applied post-DC tail = 1.836 s	Value that governed the retained full-band classification dataset
Alpha-, theta-, and C3/C4-restricted analyses	Exploratory Block 1 analyses	Applied post-DC tail = 1.836 s	Same threshold as the main full-band study
Block 1 dataset-audit analysis	Mixed export-directory report	60 metadata files and 7391 total segments, but 1740 candidate cue epochs and 1711 clean cue epochs	Useful for auditing the shared export, but not by itself the determinant of the final learning scope
Auxiliary Block 1+Block 2 modelling analysis	Supplementary run	Labelled as Block 1+Block 2, while the retained exported arrays still show Block 1 labels	Discussed in the appendix, but not used as the empirical basis of the main conclusions

Chapter 4

Methodology: Segmentation, Preprocessing, and Dataset Construction

4.1 Methodological principle

The main methodological principle of this thesis is simple: the classifier comes after a justified dataset, not before it. A segment is not a valid machine-learning sample simply because it carries a stimulus label. Artifact handling has to respect the temporal structure of the experiment rather than only local annotation names. Parameter choices also need to be described according to how they were justified: some are fixed by design, some are estimated from the data analysed here, some are carried forward from the configuration used in the final analyses, and some remain conservative defaults. For these reasons, preprocessing is treated here as a staged argument rather than a black box.

4.2 Classification of parameters by justification type

Table 4.1 summarizes the rationale categories used throughout the methodological design.

Table 4.1: Classification of key methodological parameters by type of justification.

Parameter or decision	Type of justification	Rationale
Cue-epoch target duration = 4.500 s	Design-fixed	Directly inherited from cue duration in the task
Restriction to Block 1	Scope decision	Final exported dataset and split summaries all correspond to Block 1
Stimulus tolerance = 0.300 s	Primary data-driven	Estimated from the observed distribution of cue-duration deviations in the dataset-audit analysis
Post-DC exclusion tail = 1.836 s	Signal-based estimate used in the final analyses	Adopted in the main training analysis from the recommendation generated by the DC-recovery analyses
Auxiliary DC minimum gap = 0.050 s	Conservative default	Used as a small guard threshold in the state-machine logic
Band-pass and resampling	Retained preprocessing choices	Stable signal extraction at 250.000 Hz with fixed band limits per run
STFT configuration	Retained but justified engineering choice	Read from the active configuration and held fixed across the relevant comparison
Search spaces for model hyperparameters	Conservative bounded search	Chosen for interpretable comparison under limited subject count and finite compute

4.3 Raw BrainVision loading and header-sidecar repair

Each `.vhdr` file was first read with `mne.io.read\_raw\_brainvision`. When this initial read failed, the header was repaired by parsing the `DataFile=` and `MarkerFile=` references, searching for plausible sidecar locations relative to the recording directory, the configured raw directory, and the raw-data parent directory, and then writing a repaired copy into `\_vhdr\_fixed` before repeating the read. This step is methodologically important because it shows how data loading remained robust to broken header paths without silently discarding files.

The point matters because the loading stage could not be assumed to be trivial. Even though no files were ultimately skipped in the main Block 1 analysis, explicit header repair remained part of the method and therefore belongs in the thesis description.

4.4 Signal-based estimation of DC recovery

The DC-tail recovery analysis estimates signal recovery directly from the data surrounding each detected DC marker. For each event, it extracts a fixed pre/post window around the DC onset, filters the data in the same general passband used by the main preprocessing pipeline, computes a smoothed RMS-based recovery trace, and defines recovery according to the point at which the post-DC signal falls back below a threshold derived from a prestimulus baseline period. The Block 1 recovery analysis used the following settings:

- band = 0.5–100 Hz,
- target resampling frequency = 250.000 Hz,
- pre-window = 5.000 s,
- post-window = 12.000 s,
- baseline window = $[-4.5, -0.5]$ s relative to the DC event,
- baseline percentile = 95,
- stability requirement = 0.500 s,
- RMS smoothing window = 0.050 s,
- safety margin added after the percentile summary = 0.500 s.

This stage yielded a recovery-event table, a failed-files table, a scan-summary JSON file, a valid-events CSV, a percentile-summary CSV, and four diagnostic plots: a recovery-time histogram, a boxplot, a marker-gap-versus-recovery scatter plot, and a median-RMS visualization. Together, these materials document how the recovery estimates were obtained, even though the raster images themselves are not embedded in the notebook JSON.

Table 4.2: Block 1 signal-based DC recovery summary.

Quantity	Value
VHDR files found	30
Files processed	30
Files with DC markers	29
Files without DC markers	1
Failed files	0
Total DC markers	144
Valid recovery estimates	129
Skipped: insufficient pre-window	14
Skipped: insufficient post-window	1
Median recovery	0.988 s
95th percentile recovery	1.327 s
Recommended Block 1 tail	1.827 s

Table 4.3: Comparison of the post-DC tail recommendations and the value ultimately used in the final modelling analyses.

Source	DC markers	Valid recoveries	Recommended tail
Block 1 DC-recovery analysis	144	129	1.827 s
Combined-block DC-recovery analysis	310	273	1.836 s
Final training analyses	—	—	1.836 s, used throughout classification

4.5 Boundary-based segmentation and segment ontology

The segmentation logic is boundary-based and order-aware. Rather than taking a fixed-length crop after every stimulus onset regardless of what follows, the segmentation reconstructs segments from one relevant annotation boundary to the next. This is crucial because the contamination rules depend on temporal order. A stimulus can become invalid because it is interrupted by a DC event within its own interval, or because it begins too soon after a previous DC event even when its nominal duration looks normal.

Let the ordered relevant annotations in a recording be $\{(\tau_i, d_i)\}_{i=1}^{M}$, where τ_i

is onset time and d_i is annotation description. The segmentation pass constructs ordered intervals

$$g_i = [\tau_i, \tau_{i+1}), \tag{4.1}$$

after filtering the annotation stream to the labels that matter for the final rules: task stimuli, TMS markers, DC-correction markers, and explicit new-segment boundaries. In this way, temporal context is maintained while only a subset of intervals becomes eligible for the learning dataset.

Each reconstructed interval is stored with metadata including subject, block identifier, recording identifier, source file path, within-recording order, raw annotation description, previous and next annotation descriptions, start time, end time, duration, fine label, task-candidate status, and later good/bad status. This metadata makes every retained or rejected sample auditable after model training.

4.6 Cue epoch length and the unit of rejection

The design-fixed cue-epoch length is 4.500 s. This determines the expected temporal extent of a valid anticipation sample and comes directly from the task design. It was therefore not tuned on the decoding task and was not derived from model performance.

This value matters because the aim is to classify the anticipatory state induced by the cue, not an arbitrary portion of it. A valid sample should represent the full task-defined anticipation interval. If the gap between cue onset and the next relevant boundary is much shorter than 4.500 s, that is evidence that the intended cue epoch was interrupted and cannot be fully observed.

This leads to an important methodological consequence: **the unit of rejection is the entire task stimulus epoch, not a fragment of it**. When a DC event occurs during a task stimulus, no pre-DC fragment is salvaged as a smaller sample. The entire cue epoch is rejected. Likewise, when a stimulus begins within the post-DC recovery window, the whole cue epoch is removed rather than trimming only its beginning.

4.7 Stimulus-duration tolerance

The expected cue interval lasts 4.500 s. A task segment was therefore accepted only if its duration remained close enough to that design value to count as a plausible realization of the full anticipatory window. In the final specification, the allowed

tolerance was 0.300 s, which yields an acceptable duration range from 4.200 s to 4.800 s.

This threshold was not chosen ad hoc. It was estimated in the dataset-audit analysis from the empirical distribution of task-candidate durations. Across 1740 candidate cue segments, 1725 plausible durations entered the deviation summary. The absolute deviation from the expected 4.500 s duration had a 95th percentile of 0.242 s and a 99th percentile of 0.254 s. The final threshold of 0.300 s therefore acts as a rounded, slightly conservative envelope around the observed behaviour of the data. In practical terms, it tolerates small annotation imprecision while still excluding clearly abnormal cue windows.

Table 4.4: Empirical basis of the stimulus-duration tolerance.

Quantity	Value
Task candidates examined	1740
Plausible durations used for estimation	1725
95th percentile of absolute deviation	0.242 s
99th percentile of absolute deviation	0.254 s
Selected tolerance	0.300 s
Accepted cue-duration range	4.200 s to 4.800 s

4.8 Intrinsic anomaly rules

After reconstruction, each ordered segment was checked for anomalies intrinsic to the segment itself before the broader contamination state was considered. Five cases were especially important.

1. A task cue segment shorter than the accepted range was marked invalid.
2. If such a segment ended because the next boundary was a DC event, the reason was recorded as `stim_interrupted_by_dc`.
3. If the next boundary was a TMS marker, the reason was instead recorded as `stim_interrupted_before_tms`.
4. A task cue segment longer than the accepted range was marked `stim_too_long`.
5. TMS segments themselves were never treated as machine-learning samples.

These checks formalize an important principle of dataset construction: the unit of acceptance or rejection is the full cue epoch. If a cue is interrupted before its anticipated interval is complete, no shorter fragment is salvaged as a separate sample.

The same is true for cues whose duration extends beyond what the task structure makes plausible.

More formally, the admissible duration interval was

$$\begin{aligned} \mathrm{EPOCH}_{\mathrm{MIN}} &= \mathrm{EPOCH}_{\mathrm{LEN}} - \mathrm{STIM}_{\mathrm{TOL}}, \\ \mathrm{EPOCH}_{\mathrm{MAX}} &= \mathrm{EPOCH}_{\mathrm{LEN}} + \mathrm{STIM}_{\mathrm{TOL}}. \end{aligned} \tag{4.2}$$

For the present study, $\mathrm{EPOCH}_{\mathrm{LEN}} = 4.500\,\mathrm{s}$ and $\mathrm{STIM}_{\mathrm{TOL}} = 0.300\,\mathrm{s}$, so any task segment outside [4.200 s, 4.800 s] was intrinsically invalid.

4.9 Signal-based post-DC exclusion tail

The most important contamination threshold is the post-DC exclusion tail. This determines how much time after a DC-correction marker should still be treated as potentially contaminated, even if the next cue has already begun. Two closely related signal-based estimates were obtained.

The Block 1-specific DC-recovery analysis found 144 DC markers, 129 of which yielded valid recovery estimates. Its recovery-time distribution had a 95th percentile of 1.327 s, and with the retained 0.500 s safety margin it recommended 1.827 s. The combined-block recovery analysis found 310 DC markers, 273 valid recovery estimates, a 95th percentile of 1.336 s, and a recommended value of 1.836 s. The classification analyses discussed in this thesis use 1.836 s, which is therefore treated as the operative value throughout the remainder of the manuscript.

Table 4.5: Comparison of the two signal-based DC-tail estimates carried into the modelling analyses.

Quantity	Block 1 analysis	Block 1+2 analysis
Total DC markers found	144	310
Valid DC recovery events	129	273
Median recovery	0.988 s	0.992 s
95th percentile recovery	1.327 s	1.336 s
Safety margin added	0.500 s	0.500 s
Recommended `DC\_TAIL\_S`	1.827 s	1.836 s
Used in the final main Block 1 analysis	No	Yes

The existence of both estimates is useful rather than problematic. They are close in magnitude, which supports the stability of the recovery-based logic, while 1.836 s remains the threshold that governed the Block 1 classification analyses.

4.10 Illustrative clean-versus-contaminated cue examples

The cleaning strategy is not justified by abstraction alone. Each principal modelling analysis also includes a dedicated visualization step that rebuilds one retained clean cue epoch and one rejected cue epoch directly from the segmentation state. This step is methodologically relevant because it shows how the cleaning logic translates into concrete examples rather than remaining a purely numerical argument.

The selection logic is explicit. The helper function `\_choose\_examples()` first scans the ordered segment lists and searches for a contaminated task-stimulus epoch rejected for `stim_interrupted_by_dc`. If none is available, it falls back to the first available `dc\_within\_tail` example. It then tries to find a clean epoch with the same fine label as the contaminated example so that the visual comparison is not confounded by class identity. Only if no label-matched clean epoch exists does it fall back to any retained clean task epoch. In other words, the figure selection is neither random nor retrospectively hand-picked. It follows a fixed priority rule that prefers the contamination mechanism most directly tied to whole-epoch interruption and then attempts to keep the task label constant across the clean-versus-contaminated comparison.

Once an example is chosen, the original BrainVision file is reopened, the full 4.500 s stimulus window is extracted from the raw signal, the data are resampled to the thesis sampling rate, and a small set of representative channels is plotted. The channel-selection helper uses a fixed preference list — `Fz`, `Cz`, `Pz`, `Oz`, `Fp1`, `Fp2`, `C3`, `C4`, `P3`, and `P4` — and keeps up to six channels in the full-channel analyses. The C3/C4-only analysis reuses the same visualization machinery but limits the retained signal itself to those two channels. Dashed vertical lines are added for DC markers that occur inside the plotted stimulus window, and the summary CSV written next to the figure stores the example type, subject, block, recording identifier, file path, within-recording order, fine label, rejection reason, epoch duration, number of DC markers inside the displayed window, and the delay from the previous DC marker when available.

Each modelling analysis therefore produced a representative clean-versus-contaminated example figure together with a compact summary table describing the selected epochs. These figures are not incidental debugging traces. They provide direct visual support for the claim that the rejected cue epochs are not simply unusual durations, but concrete examples of cue windows that are interrupted by

DC or that begin before post-DC recovery has elapsed.

4.11 Order-aware state machine for contamination handling

Contamination handling is implemented through a state machine over the ordered segment sequence. A task epoch is invalid not only when a DC event occurs inside it, but also when it begins before the signal has recovered from a preceding DC event.

Let t_i^{start} be the onset time of candidate stimulus epoch i, and let $t_{\text{last DC}}$ be the onset of the most recent DC event preceding that epoch in the ordered stream. The state machine rejects the entire stimulus epoch if

$$0 \leq t_i^{\text{start}} - t_{\text{last DC}} < \text{DC_TAIL}. \tag{4.3}$$

In the present study, DC_TAIL = 1.836 s, and the rejection reason is stored as `dc\_within\_tail`. This rule directly addresses the supervisors' concern that annotation boundaries alone may not fully capture signal usability.

A second whole-epoch rule concerns interruption. If a task stimulus begins and a DC marker occurs before the cue interval is complete, the resulting boundary-to-boundary task segment is too short. In that case, the reason `stim_interrupted_by_dc` invalidates the entire cue epoch. Again, no part of the interrupted interval is salvaged as a separate learning sample.

4.12 Auxiliary DC minimum gap (`DC\_MIN\_S`)

A smaller parameter appearing in the code is `DC\_MIN\_S = 0.050`. Unlike `STIM\_TOL\_S` and `DC\_TAIL\_S`, this threshold is not one of the primary data-driven scientific results of the pipeline. It is better described as a conservative auxiliary default. A threshold of 0.050 s is small enough to catch clearly degenerate marker gaps without materially determining which normal cue epochs survive dataset construction.

4.13 Filtering and resampling

After a task epoch passed the segmentation and contamination checks, the signal extraction stage applied a Butterworth band-pass filter and resampled the retained

segment to a common sampling rate of 250.000 Hz. In the main full-band and C3/C4 runs, the passband was 0.5–100 Hz with filter order 4. In the alpha-only and theta-only variants, the same extraction logic was reused but the passband was restricted to 8–12 Hz and 4–8 Hz respectively.

These filter settings are best described as retained preprocessing choices rather than newly optimized thresholds. Their role was to provide stable and comparable extraction while preserving the frequency content most relevant to the present question and attenuating very slow drift and higher-frequency noise that were not central to the analysis [2, 3]. Resampling to 250.000 Hz harmonized all retained epochs to a common temporal resolution and yielded a fixed cue length of 1125 samples for a 4.500 s epoch.

The exact implementation matters. Filtering is performed on the segment data after extraction from the raw object. The helper function `\_safe\_sos\_filter` constructs a fourth-order Butterworth band-pass in second-order-sections form. When the segment is long enough, the code uses `sosfiltfilt` to obtain a zero-phase result; if that is not possible, it falls back to `sosfilt`. Resampling is handled by `\_resample\_exact`: if the ratio between the input and target sampling rates is an integer decimation factor, the code uses `signal.decimate(..., zero\_phase=True)`; otherwise it uses `signal.resample\_poly`. This detail is worth reporting because it clarifies why the preprocessing was more careful than a generic "filter and resample" summary might suggest.

4.14 Extraction of fixed-size time-domain tensors

Let $x_c(t)$ denote the signal from channel c. After filtering and resampling, the preprocessing stage padded or truncated each retained epoch to a fixed length corresponding to 4.500 s at 250.000 Hz, namely 1125 time points. The code therefore exported a time-domain tensor

$$\mathbf{X}^{\text{time}} \in \mathbb{R}^{N \times C \times T}, \tag{4.4}$$

where $N = 1711$, $T = 1125$, and $C = 64$ in the full-band, alpha-only, and theta-only runs, while $C = 2$ in the C3/C4-only run.

4.15 STFT representation and retained settings

The time–frequency dataset is derived from exactly the same cleaned epochs. The STFT configuration loaded by the main Block 1 analysis was:

- Hann window,
- `nperseg` corresponding to 3.000 s, that is 750 samples at 250.000 Hz,
- overlap corresponding to 2.800 s, that is 700 samples,
- `nfft` equal to four times `nperseg`, namely 3000,
- frequency truncation at 40.000 Hz.

These values were held fixed within the relevant comparison. In the main analysis, the STFT tensor had shape $(1711, 64, 481, 8)$, with 481 frequency bins spanning 0–40 Hz and eight time bins whose saved coordinates ranged from approximately 1.500 s to 2.900 s. In the C3/C4-only restriction, the corresponding shape was $(1711, 2, 481, 8)$.

4.16 Exported outputs and provenance records

Beyond the model-ready arrays, the study produced a set of outputs that can be grouped into six categories:

1. cleaned time-domain and time-frequency datasets for downstream modelling;
2. dataset-construction summaries describing how many segments were scanned, rejected, and retained;
3. per-recording metadata tables carrying subject, recording, order, label, timing, and quality-control information;
4. split manifests and normalization statistics derived from the training partition only;
5. search summaries, selected-model reports, and checkpoint metadata for each model family; and
6. diagnostic figures such as class-balance plots, learning curves, compact validation-versus-test summaries, and confusion matrices.

Taken together, these outputs make it possible to relate the final classification results back to the dataset-construction process that produced them. A retained

sample can still be traced to its recording context, and the summary tables reported in the thesis remain linked to records generated directly during the study.

4.17 Diagnostic reports and figure inventory

In addition to the final arrays, the study also generated a substantial set of diagnostic reports. For the full-band analysis, and again for the alpha, theta, and C3/C4 variants, this included split-distribution summaries, class-balance diagnostics, hyperparameter-search tables, compact held-out summary plots, train/validation learning curves, learning-rate traces, confusion matrices, and plain-text classification reports for the selected models.

These materials matter because they provide a much richer visual record than a small set of headline result tables alone would suggest. They are not peripheral to the interpretation of the results. The main comparisons are discussed in the body of the thesis, while additional supplementary figures are gathered in Appendix D. Table 4.6 summarizes the main kinds of material generated during the study.

Table 4.6: Main categories of analytical output used throughout the thesis.

Output category	Typical contents	Use in the thesis
DC-recovery diagnostics	Recovery-time distributions, marker-gap summaries, and threshold notes	Motivate and justify the post-DC exclusion tail
Dataset-audit summaries	Duration histograms, removal summaries, and per-recording counts	Support the cleaned-dataset audit and the stimulus-tolerance discussion
Illustrative quality-control figures	Representative clean and contaminated cue examples	Show how the contamination rules map onto concrete signal traces
Split and balance reports	Partition counts, class distributions, and sampler summaries	Document the grouped split and the near-balanced class structure
Training diagnostics	Loss, accuracy, macro-F1, learning-rate, and confusion-matrix figures	Describe the optimization behaviour of the selected models
Compact result summaries	Validation-versus-test comparison plots and summary tables	Provide a concise overview of setting-specific performance

4.18 Result of the cleaning stage

The Block 1 dataset-audit analysis showed that, after all rules were applied, 29 task epochs were removed. Importantly, these removals were concentrated in exactly the two DC-related categories shown in Table 4.7.

Table 4.7: Task-epoch removals by rule in the executed Block 1 dataset-audit analysis.

Removal reason	Count
`stim_interrupted_by_dc`	16
`dc\_within\_tail`	13
Total removed task epochs	29

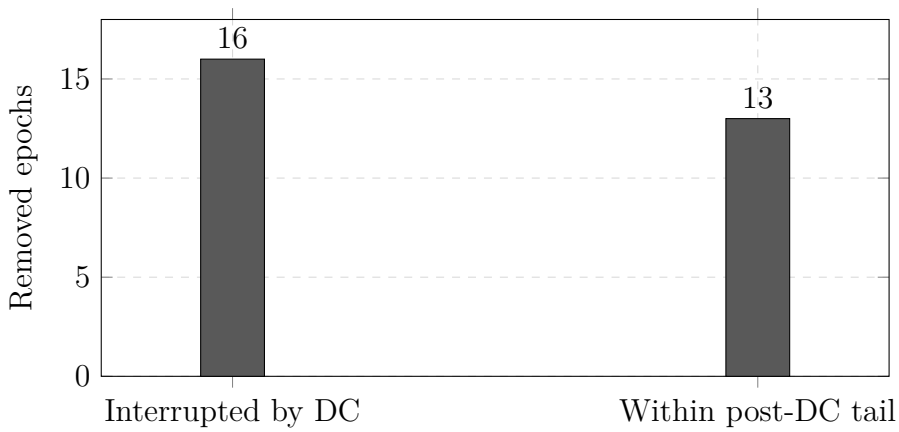

Figure 4.1: Removed task epochs by rule. The removals were concentrated in the exact DC-related mechanisms that motivated the contamination-control strategy.

Because the removals were targeted rather than diffuse, the cleaning stage can be interpreted as conceptually specific rather than broadly punitive.

Chapter 5

Representations, Models, Training Procedure, and Evaluation Protocol

5.1 Time-domain and time–frequency inputs

The main Block 1 run exported two matched representations from the same cleaned source data.

Time domain. The raw model input is the multichannel signal tensor of shape (N, C, T). In the main run, $N = 1711$, $C = 64$, and $T = 1125$. This representation preserves waveform structure directly and leaves feature extraction to the neural network.

Time–frequency domain. The STFT representation encodes each epoch as a channel-by-frequency-by-time tensor. In the full-band main run, the shape is $(1711, 64, 481, 8)$. This makes oscillatory structure explicit while compressing the temporal axis into a small number of windows.

The methodological value of this pairing is that it separates representation from cleaning. Any performance difference between the time-domain and STFT-based models cannot be attributed to different retained sample sets in the reported run.

5.2 Grouped split, safety fallback, and train-only normalization

The data were partitioned subject-wise with repeated `GroupShuffleSplit` attempts until a valid split satisfying a minimum-per-class condition was found. In the present study, the resulting split mode was `grouped\_ok(seed=42,min\_per\_class=2)`. The intended grouped split was therefore achieved without resorting to the sample-wise fallback path kept in reserve as a safety mechanism.

The final partition sizes were:

- training: 1117 samples from 19 subjects,
- validation: 296 samples from 5 subjects,
- test: 298 samples from 5 subjects.

There was zero overlap of subjects and zero overlap of recording identifiers across train, validation, and test.

Normalization was computed on the training partition only and then applied to validation and test. For the time-domain tensor, the normalized signal takes the generic form

$$\hat{x}_{nct} = \frac{x_{nct} - \mu_c^{\text{train}}}{\sigma_c^{\text{train}} + \varepsilon}, \tag{5.1}$$

where the mean and standard deviation are estimated exclusively from the training data. The same principle was applied to the spectrogram representation. This prevents leakage of distributional information from held-out subjects into development.

Table 5.1: Grouped split summary for the final main Block 1 analysis.

Partition	Samples	Unique subjects	Class counts
Train	1117	19	373 left-threat, 372 right-threat, 372 safety
Validation	296	5	99 left-threat, 100 right-threat, 97 safety
Test	298	5	99 left-threat, 100 right-threat, 99 safety

5.3 Weighted sampler and class balance

Training used a weighted random sampler built from inverse class frequency on the training partition only. In practice, however, the training set was already almost perfectly balanced: the class-balance report gave 373, 372, and 372 samples across the three classes, with a max/min non-zero imbalance ratio of approximately 1.0027. The weighted sampler should therefore be interpreted as a retained training-stabilization choice rather than as a substantive rebalancing intervention.

5.4 Training procedure

All neural models were trained under the same broad training logic, although the time-domain and time–frequency branches differ slightly in early-stopping patience

and in the batch-size search range. The random seed was fixed at 42 for Python, NumPy, and PyTorch, and deterministic CuDNN behaviour was enabled where available. Training was carried out on CPU. Optimization used AdamW with the learning rate and weight decay defined by the current hyperparameter configuration. The loss was cross-entropy with label smoothing 0.05. Validation macro-F1 was the sole model-selection signal.

The common training loop was:

1. construct a weighted-sampler training loader and deterministic validation/test loaders;
2. train for up to 80 epochs;
3. after each epoch, evaluate on the validation set and append train loss, validation loss, train accuracy, validation accuracy, validation macro-F1, and the current learning rate to the history object;
4. pass validation macro-F1 to a *ReduceLROnPlateau* scheduler configured in maximization mode, with reduction factor 0.5 and patience 3;
5. save the best model state whenever validation macro-F1 improves by more than 10^{-4};
6. stop if the number of non-improving epochs reaches the patience threshold; and
7. reload the best validation state before the final validation and test evaluation.

The patience threshold was 12 epochs for the time-domain TCN families and 10 epochs for the spectrogram CNN family. During backpropagation, gradients were clipped to a maximum ℓ_2 norm of 1.0. Each trial retained the best validation state as a checkpoint, and the search stage also generated summary tables, learning curves, confusion matrices, plain-text classification reports, and a compact JSON summary of the selected models.

Table 5.2: Training-loop settings used in the modelling analyses.

Setting	Value used in the runs
Random seed	42 for Python, NumPy, and PyTorch
Device used in this study	CPU
Optimizer	AdamW
Learning-rate schedule	Reduce-on-plateau schedule on validation macro-F1 (factor 0.5, patience 3)
Maximum epochs per trial	80
Early-stopping patience	12 epochs for time-domain models; 10 epochs for spectrogram models
Gradient clipping	Maximum norm of 1.0
Model-selection signal	Validation macro-F1
Checkpointing	Best validation state retained for each trial
Outputs produced after search	Search tables, summary JSON, classification reports, learning curves, and confusion matrices

5.5 Loss function and evaluation metrics

The models were trained with a cross-entropy objective using label smoothing of 0.05. In generic form, the smoothed target distribution for class k among K classes can be written as

$$q_i = \begin{cases} 1 - \alpha + \frac{\alpha}{K}, & i = k, \\ \frac{\alpha}{K}, & i \neq k, \end{cases} \tag{5.2}$$

with $\alpha = 0.05$ and $K = 3$ in the present study.

Model selection was based on validation macro-F1, while accuracy was also monitored and reported. Macro-F1 was chosen as the main selection metric because it weights classes equally and is less likely than plain accuracy to hide class-specific degradation in a three-class problem:

$$\text{Macro-F1} = \frac{1}{K} \sum_{k=1}^{K} \text{F1}_k. \tag{5.3}$$

The test set was evaluated only for the final selected model within each family.

5.6 Baseline TCN architecture

The time-domain baseline model is a temporal convolutional network with three dilated convolutional stages. It applies input batch normalization, then three `Conv1d` blocks with kernel sizes drawn from the current configuration, batch normalization, GELU activations, dilation factors 1, 2, and 4, and `MaxPool1d(2)` after each stage. The backbone is followed by adaptive average pooling, dropout, a small fully connected hidden layer, and the final classifier.

The best full-band baseline trial used channel widths $(64, 96, 128)$, kernel sizes $(9, 7, 5)$, dropout 0.2, learning rate 10^{-4}, weight decay 2×10^{-4}, and batch size 8.

5.7 Attention-augmented TCN

The second time-domain family extends the baseline with a learned temporal attention-pooling layer. After the convolutional backbone produces a feature sequence $\mathbf{h}_t$, the attention module computes scalar scores over time, applies a softmax, and forms a weighted pooled vector:

$$\alpha_t = \frac{\exp(s_t)}{\sum_j \exp(s_j)}, \qquad \mathbf{z} = \sum_t \alpha_t \mathbf{h}_t. \tag{5.4}$$

This allows the model to emphasize the most informative parts of the cue interval while remaining close to the baseline architecture.

The best full-band attention TCN used channel widths $(64, 96, 128)$, kernel sizes $(11, 9, 7)$, dropout 0.2, attention hidden size 32, learning rate 10^{-4}, weight decay 10^{-4}, and batch size 8.

5.8 Spectrogram CNN

The time–frequency family uses a compact convolutional neural network over spectrograms. It contains three two-dimensional convolutional blocks with batch normalization and GELU activations. Pooling is applied only along the frequency axis, with a pooling kernel of 2 and stride 2 in frequency while leaving the time axis unchanged. In this way the model preserves the small number of STFT time windows while progressively aggregating spectral structure. The backbone is followed by adaptive average pooling, dropout, and a final linear classifier.

The best full-band spectrogram CNN used base channels 32, dropout 0.3, learning

rate 3×10^{-4}, weight decay 10^{-4}, and batch size 32.

5.9 Architecture detail and parameter counts

Because the analysis records preserve parameter counts for every evaluated trial, the thesis can report model capacity directly rather than only qualitatively. In the main full-band run, the selected baseline TCN contained 150,755 trainable parameters, the selected attention TCN 199,972, and the selected spectrogram CNN 111,651. In the C3/C4-only variant the parameter counts changed because the number of input channels was reduced to two: the selected baseline TCN contained 114,919 parameters, the selected attention TCN 154,120, and the selected spectrogram CNN 23,859.

These counts matter because they show that the attention model's advantage in the main run was not obtained by an order-of-magnitude increase in capacity. They also clarify that the C3/C4-only restriction changes not only the input representation but also the parameterization of the early layers.

Table 5.3: Parameter counts of the selected models in the present study.

Setting	Model family	Trainable parameters
Full-band	Baseline TCN	150,755
Full-band	Attention TCN	199,972
Full-band	Spectrogram CNN	111,651
Alpha-only	Baseline TCN	89,475
Alpha-only	Attention TCN	70,564
Alpha-only	Spectrogram CNN	111,651
Theta-only	Baseline TCN	68,995
Theta-only	Attention TCN	199,972
Theta-only	Spectrogram CNN	111,651
C3/C4-only	Baseline TCN	114,919
C3/C4-only	Attention TCN	154,120
C3/C4-only	Spectrogram CNN	23,859

5.10 Search-space design

The modelling procedure relied on bounded grid-based search spaces and then sampled manageable subsets from the full Cartesian grids. For the time-domain

families, the full baseline grid contained 32 configurations and the full attention grid 64 configurations; the study evaluated 12 baseline trials and 16 attention trials. For the spectrogram family, the full grid also contained 32 configurations and the study evaluated 16 sampled trials.

These ranges were not meant to exhaust the full model-design space. Their purpose was to support a restrained and interpretable comparison under fixed preprocessing and a limited number of subjects. Tables 5.4 and 5.5 summarize the selected ranges.

Table 5.4: Time-domain hyperparameter search space used in this study.

Parameter	Values explored
Channels	$(32, 64, 96)$, $(64, 96, 128)$
Kernels	$(9, 7, 5)$, $(11, 9, 7)$
Dropout	0.20, 0.30
Learning rate	10^{-4}, 3×10^{-4}
Weight decay	10^{-4}, 2×10^{-4}
Batch size	8
Label smoothing	0.05
Attention hidden size	16 or 32 (attention model only)

Table 5.5: Time–frequency hyperparameter search space used in this study.

Parameter	Values explored
Base channels	16, 32
Dropout	0.20, 0.30
Learning rate	10^{-4}, 3×10^{-4}
Weight decay	10^{-4}, 10^{-3}
Batch size	16, 32
Label smoothing	0.05

5.11 Exploratory variants

Beyond the main full-band analysis, the thesis preserves three targeted exploratory reruns.

Alpha-only variant. The extraction pipeline was rerun with an 8–12 Hz band-pass filter while keeping the same segmentation logic, class mapping, grouped split sizes, and bounded search structure.

Theta-only variant. The extraction pipeline was rerun with a 4–8 Hz band-pass filter. Unlike the older thesis draft, the theta analysis includes time-domain baseline and attention models as well as the spectrogram CNN, so the thesis reports all three theta families.

C3/C4-only variant. The pipeline was rerun while retaining only channels C3 and C4, producing tensors of shape $(1711, 2, 1125)$ in the time domain and $(1711, 2, 481, 8)$ in the STFT domain.

The exploratory variants are informative but remain secondary to the main full-band analysis. They change the input restriction while holding the broader cleaning and evaluation logic fixed.

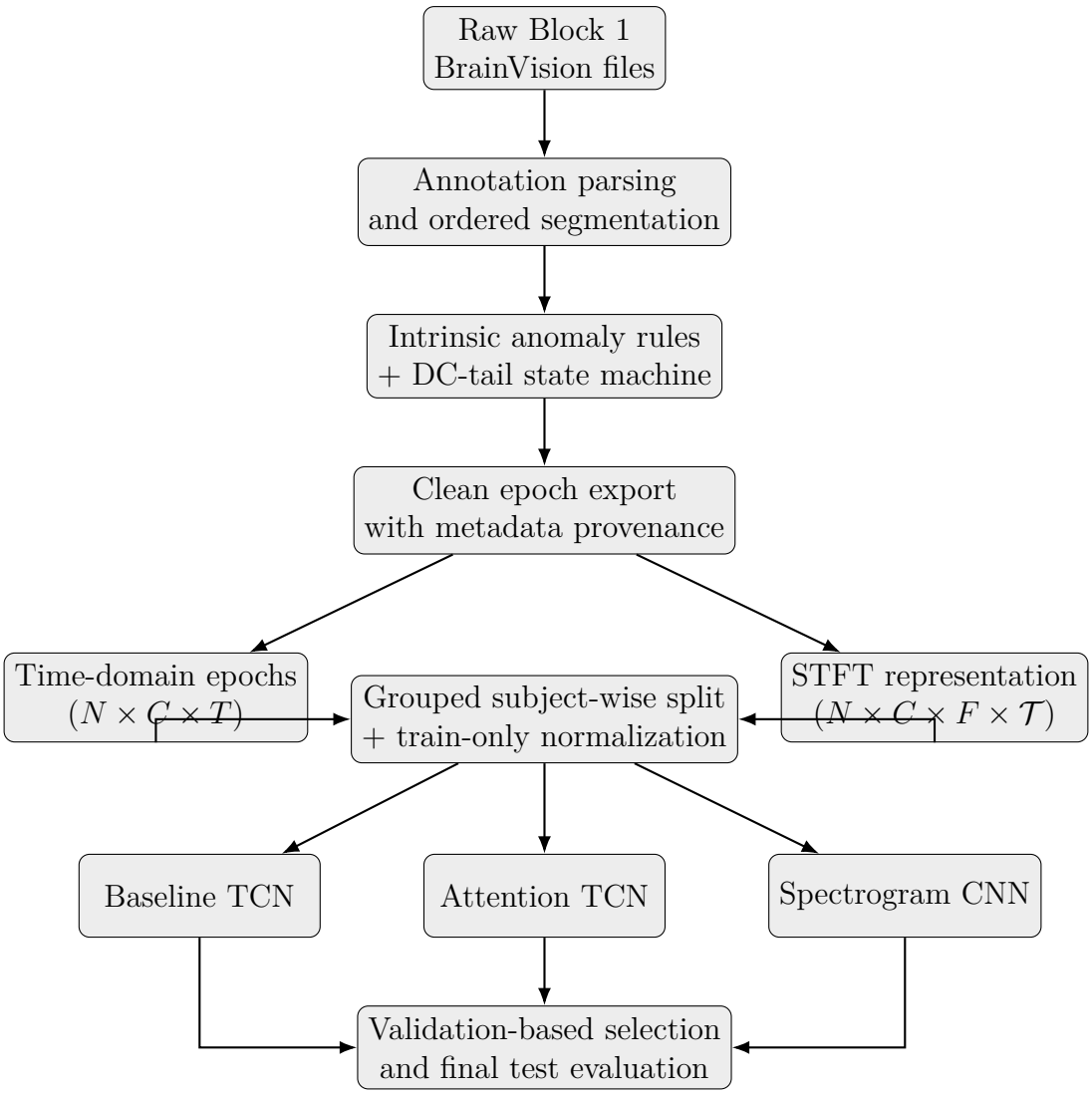

Figure 5.1: Overview of the present study pipeline. The classifier stage is the final step in a longer chain of justified transformations.

Chapter 6

Results

6.1 Quality and balance of the final exported dataset

The cleaned Block 1 export contained 1711 epochs and was almost perfectly balanced across the three classes. This is already an important result of the pipeline, because it shows that contamination control did not distort the class distribution. Starting from 1740 task candidates and removing only 29 invalid epochs, the final class counts remained 571 left-threat, 572 right-threat, and 568 safety. As a result, the later model comparison is not confounded by strong class imbalance.

It is also important that the grouped split maintained this balance across train, validation, and test. The training partition contains 373, 372, and 372 samples across the three classes; the validation partition contains 99, 100, and 97; and the test partition contains 99, 100, and 99.

6.2 Main full-band decoding results

Table 6.1 reports the main results from the executed full-band Block 1 run. Two points stand out.

First, the attention-augmented TCN achieved the best held-out test performance among the full-band models, reaching 39.93% accuracy and macro-F1 0.3867. Second, the spectrogram CNN achieved the strongest validation performance, but its test performance dropped to 35.23% accuracy and macro-F1 0.3535. This validation–test gap suggests that an explicit time–frequency representation did not automatically improve subject-generalizing decoding in this study.

For a balanced three-class problem, chance accuracy is 33.33%. The best full-band model therefore exceeded chance by roughly 6.6 percentage points on the held-out test set. This improvement is modest rather than dramatic, but it remains meaningful precisely because it was obtained under a grouped subject-wise split with train-only

Table 6.1: Main full-band Block 1 results from the present study.

Model family	Val. macro-F1	Val. acc. (%)	Test macro-F1	Test acc. (%)
Baseline TCN	0.3950	39.53	0.3637	36.58
Attention TCN	0.3839	38.85	**0.3867**	**39.93**
Spectrogram CNN	**0.4253**	**42.57**	0.3535	35.23

normalization. In the context of EEG decoding, that conclusion is strengthened by evaluation safeguards designed specifically to reduce optimistic bias under limited sample sizes [17, 26]. It is also consistent with the broader pain-prediction literature in which pre-stimulus and anticipatory signals can be informative without making the decoding problem trivial [35, 36].

6.3 Summary figures for the main full-band comparison

To complement the numerical comparison in Table 6.1, the main full-band analysis produced compact bar plots comparing validation and test performance for the selected time-domain and time–frequency models. Figure 6.1 brings these panels together: validation-versus-test accuracy and validation-versus-test macro-F1 for the selected TCN models, alongside the corresponding pair for the selected spectrogram model.

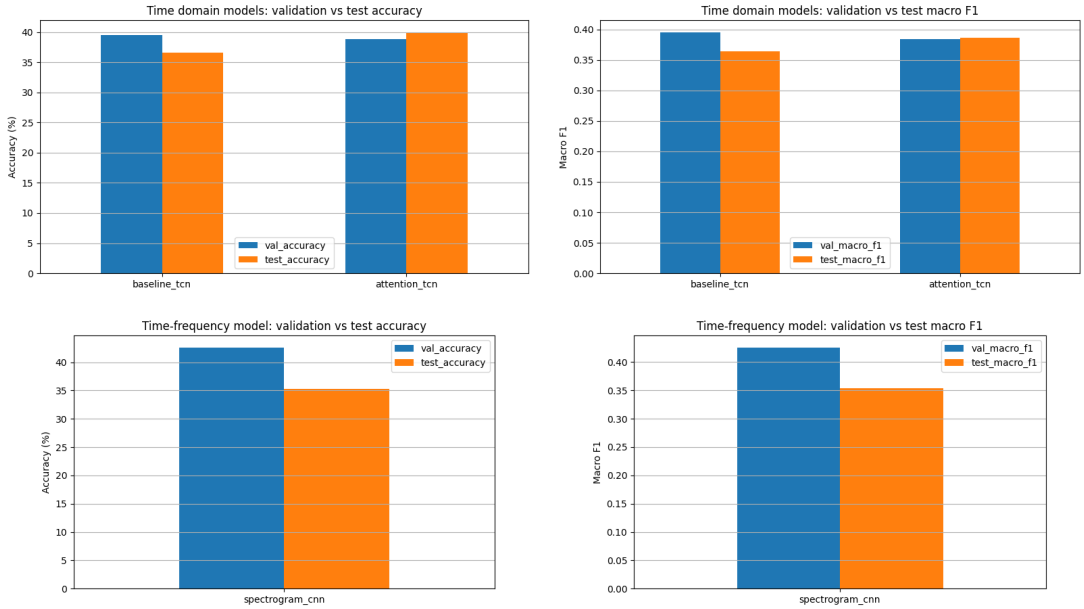

Figure 6.1: Compact summary figures for the main full-band analysis. The upper row shows the validation-versus-test comparison for the selected baseline and attention TCN models in terms of accuracy and macro-F1. The lower row shows the corresponding validation-versus-test comparison for the selected spectrogram CNN. Together, these panels provide a compact visual summary of the full-band model comparison.

The most important contrast visible in Figure 6.1 is between the time-domain and spectrogram models. The attention TCN shows the most stable validation-to-test behaviour, whereas the spectrogram CNN displays the largest drop from validation to test.

6.4 Learning curves and confusion matrices of the selected full-band models

The scalar scores in Table 6.1 are further contextualized by the training diagnostics shown in Figures 6.2 and 6.3. Figure 6.2 presents the time-domain diagnostics for the selected baseline and attention TCN models: train/validation loss, train/validation accuracy, validation macro-F1 over epochs, learning-rate evolution under the scheduler, and the final test confusion matrix for each model. Figure 6.3 shows the corresponding diagnostics for the selected spectrogram CNN.

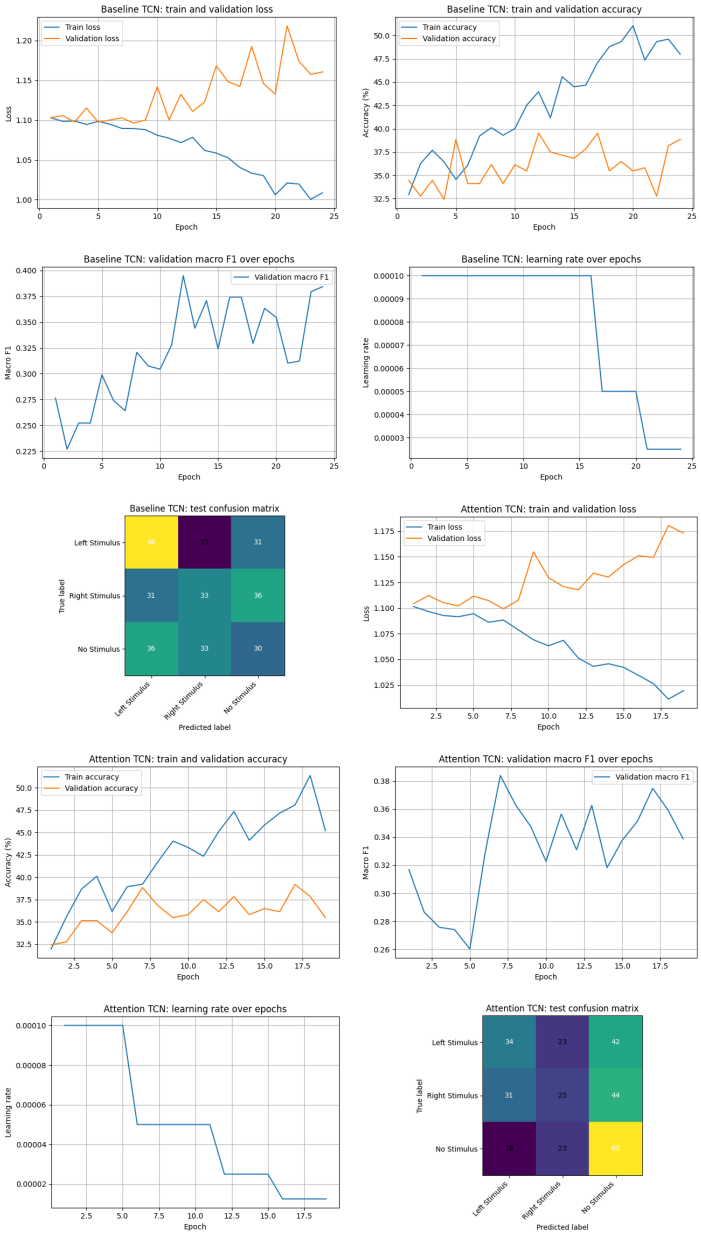

Figure 6.2: Time-domain diagnostic figures for the main full-band analysis. The figure reproduces, for the selected baseline and attention TCN models, the train/validation loss curves, train/validation accuracy curves, validation macro-F1 curves, scheduler-driven learning-rate curves, and the final test confusion matrices.

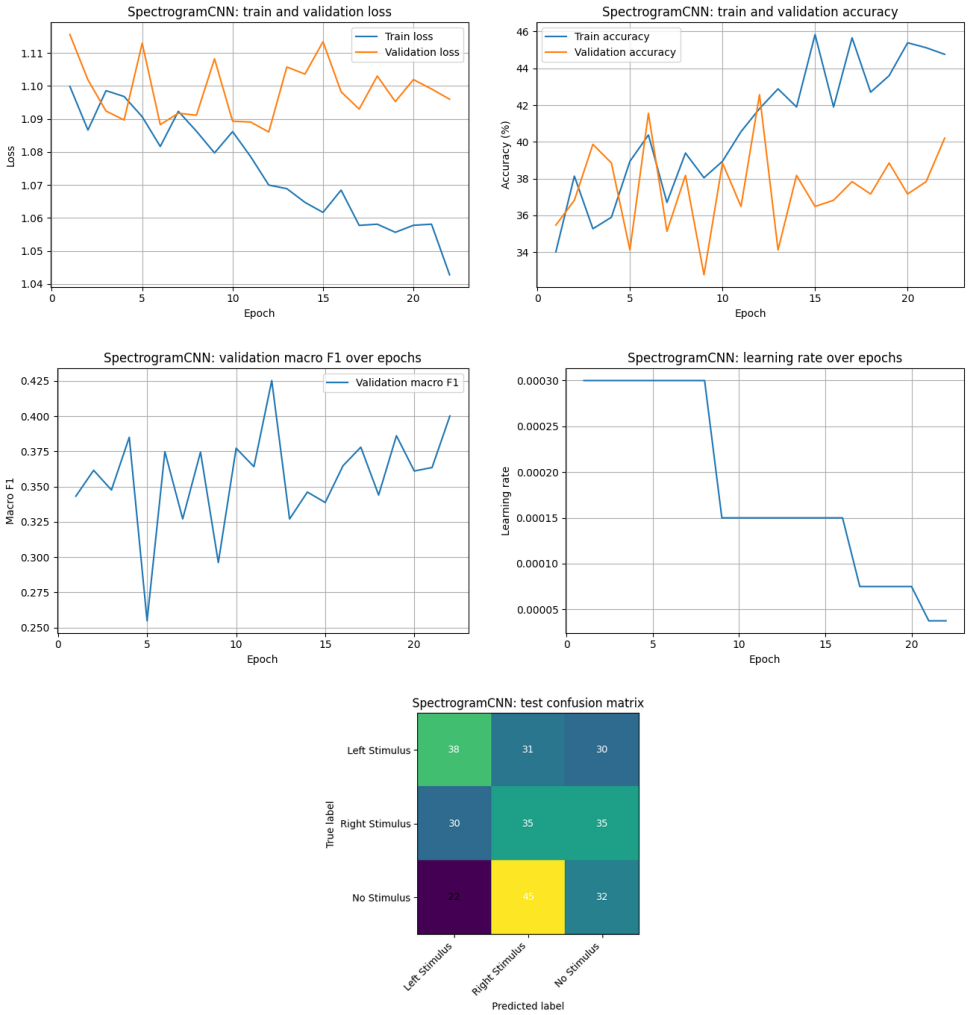

Figure 6.3: Time-frequency diagnostic figures for the main full-band analysis. The figure reproduces the train/validation loss, train/validation accuracy, validation macro-F1, learning-rate, and test confusion-matrix plots for the selected spectrogram CNN.

The confusion matrices make the class-specific behaviour more concrete. In the selected attention TCN, the safety class is recognized most strongly, whereas left-threat and right-threat trials are more frequently confused with safety than with one another. The baseline TCN shows a more even but weaker class distribution, and the selected spectrogram CNN exhibits a pronounced tendency to misclassify safety epochs as right-threat. These qualitative patterns are consistent with the aggregate macro-F1 and accuracy values reported in Table 6.1: the models are extracting real structure, but the class boundaries remain modest rather than sharp.

6.5 Validation-to-test generalization behaviour

A direct comparison of validation and test performance is also informative. The baseline TCN shows a moderate drop from validation to test, the attention TCN is comparatively stable and even improves slightly in accuracy, and the spectrogram CNN shows the strongest deterioration from validation to test. Table 6.2 summarizes this pattern.

Table 6.2: Validation-to-test change for the main full-band models. Positive values indicate improvement from validation to test.

Model family	Δ Accuracy (test – val, pp)	Δ Macro-F1 (test – val)
Baseline TCN	-2.95	-0.0313
Attention TCN	+1.08	+0.0027
Spectrogram CNN	-7.34	-0.0718

The attention-based temporal summarization was therefore the most stable of the three main model families on unseen subjects in the full-band analysis. A cautious interpretation is that the attention mechanism helped the model emphasize the most informative portions of the cue interval without forcing the signal into an aggressively compressed time–frequency summary. In a subtle subject-wise decoding problem, that balance may be advantageous: it preserves temporal detail while still allowing the network to weight informative segments adaptively. By contrast, the larger validation-to-test drop in the spectrogram CNN suggests that the compact STFT representation may have captured regularities that were useful within the validation split but less robust across unseen participants. Read together with the broader project results, this pattern suggests that preserving the temporal evolution of cue-period activity may be especially important when the signal of interest includes both sustained threat expectancy and more spatially specific preparatory dynamics [4, 5, 40, 41].

6.6 Exploratory variant results

The exploratory analyses broaden the interpretation of the main result by asking whether decoding changes when the signal is restricted to a theoretically motivated frequency band or to a narrow central-channel subset. Because these analyses address slightly different questions, their results are easier to read when presented as three smaller tables rather than as one dense summary.

Table 6.3: Results for the alpha-restricted analysis.

Model family	Val. macro-F1	Val. acc. (%)	Test macro-F1	Test acc. (%)
Baseline TCN	0.4297	42.91	0.3448	35.23
Attention TCN	0.4121	41.22	**0.3975**	**40.27**
Spectrogram CNN	0.3705	37.50	0.3047	30.87

Table 6.4: Results for the theta-restricted analysis.

Model family	Val. macro-F1	Val. acc. (%)	Test macro-F1	Test acc. (%)
Baseline TCN	0.3625	36.49	0.2980	31.54
Attention TCN	0.3859	38.85	0.3678	36.91
Spectrogram CNN	0.3978	40.54	0.3574	37.25

Table 6.5: Results for the C3/C4-restricted analysis.

Model family	Val. macro-F1	Val. acc. (%)	Test macro-F1	Test acc. (%)
Baseline TCN	0.3615	36.15	0.3616	36.58
Attention TCN	0.3632	37.16	0.2808	33.22
Spectrogram CNN	0.3897	39.19	0.3281	33.22

Three observations are especially noteworthy.

Alpha-only attention performs best overall. The alpha-only attention TCN achieved 40.27% test accuracy and macro-F1 0.3975, slightly stronger than the best full-band model. This fits well with the broader project interpretation that somatomotor alpha carries especially body-specific anticipatory information [7, 8, 9, 4, 40, 41]. It also sits naturally alongside the project-level finding that alpha predicted MEP modulation, suggesting that alpha-band dynamics may capture a preparatory state closely related to the limb-specific motor adjustments reported in the wider study [40, 41].

Theta-only remains informative but not dominant. The best theta result came from the spectrogram CNN at 37.25% test accuracy and macro-F1 0.3574, closely followed by the theta attention TCN at 36.91% and macro-F1 0.3678. Theta therefore retains useful information, but it did not surpass the alpha-only attention result [4, 5, 23, 24, 40, 41]. This is compatible with a role for theta in broader threat valuation and updating rather than in the most spatially specific aspect of anticipation.

C3/C4-only is too restrictive to be optimal. Restricting the input to two

central channels did not improve overall performance. The best C3/C4 result only matched the full-band baseline accuracy, and the C3/C4 attention model degraded noticeably on the test set. That pattern suggests that although sensorimotor channels are relevant, the decodable information is not reducible to a single narrowly local motor signature and likely benefits from a wider anticipatory network that includes both lateralized and more distributed contributions.

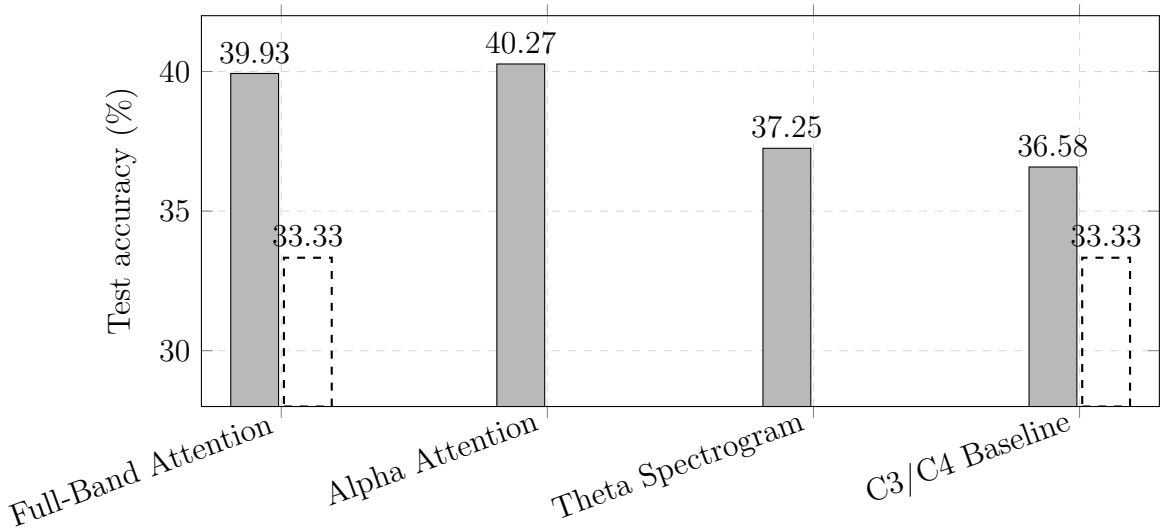

Figure 6.4: Held-out test accuracy across selected main and exploratory settings. The dashed reference indicates balanced three-class chance accuracy.

6.7 What the diagnostic figures add beyond the headline metrics

The compact bar plots and learning diagnostics are not decorative additions. They document how the chosen models arrived at the reported held-out scores. In the main full-band analysis, the time-domain diagnostic grid shows that both TCN families stopped well before the hard limit of 80 epochs, consistent with the implemented patience-based early-stopping rule. The learning-rate plots also make the scheduler behaviour visible: once validation macro-F1 plateaued, the learning rate was halved in discrete steps, exactly as specified by the `ReduceLROnPlateau` schedule with factor 0.5 and patience 3. The same pattern appears in the spectrogram diagnostic figures, although the time-frequency branch typically runs for a somewhat different number of epochs because it uses its own patience threshold of 10 epochs rather than 12.

The confusion matrices matter as much as the scalar metrics. In the main attention-TCN result, the strongest diagonal entry is the safety class, while the two

threat-direction classes are still frequently confused with safety and with each other. This is consistent with a task that is decodable but not trivially separable. The baseline TCN shows a similar general structure but with a weaker diagonal overall. The spectrogram CNN likewise remains above chance overall, yet its test confusion matrix reflects the same broad limitation already visible in the validation-to-test drop: the model does not preserve its validation advantage when confronted with unseen subjects. The alpha-, theta-, C3/C4-, and auxiliary Block 1+Block 2 analyses use the same style of diagnostic figures, and those grids are gathered in Appendix D so that the surrounding prose can be read against the underlying visual evidence.

For that reason, the thesis treats the diagnostic figures as part of the results, not merely as optional appendices. The bar plots show whether validation-selected models remain competitive on the held-out test set. The training curves show whether those scores were reached through stable or unstable optimisation trajectories. The confusion matrices show which classes benefit most from each representation and which confusions remain unresolved. Taken together, they make the numerical summaries easier to interpret and help prevent the discussion from overstating what the classifier has learned.

6.8 Answer to the research question

Taken strictly on the evidence assembled here, the answer to the thesis question is yes, but with appropriate caution. Under the Block 1-only, grouped subject-wise, train-only-normalized protocol, the cleaned cue-period EEG contains enough information for models to distinguish left-threat, right-threat, and safety cue conditions at above-chance accuracy. The strongest confirmatory full-band result is the attention TCN with test accuracy 39.93% and macro-F1 0.3867. The alpha-only exploratory result further supports the idea that body-specific information is present and may be especially concentrated in somatomotor alpha-band dynamics.

At the same time, the performance remains moderate. This is not a case in which classification becomes easy once the right architecture is applied. The final results are better understood as evidence that spatially specific anticipatory information is recoverable but limited in the available EEG signal.

6.9 What is not claimed by these results

The results do not establish statistical significance against a null distribution, because permutation-based confidence intervals were not computed for the decoding metrics. They also do not show that the best model architecture is universally superior, because the dataset is modest and the search space was intentionally bounded. Finally, the numerical findings do not automatically generalize to Block 2 reversal, since that phase was excluded from the learning dataset used for the present study.

6.10 Auxiliary Block 1+Block 2 analysis

The thesis also documents an analysis labelled as a Block 1+Block 2 run. Although the initial scan covered 30 Block 1 files and 30 Block 2 files, the later split summary still reports only blocks `['1']` in the exported arrays used for learning. This auxiliary Block 1+Block 2 analysis is therefore discussed as supplementary evidence rather than as the basis of a jointly trained two-block classifier.

Even so, the auxiliary Block 1+Block 2 analysis remains informative because it shows how the same modelling framework behaves outside the principal Block 1 study. Its best attention-TCN result reached 42.95% test accuracy with macro-F1 0.4239, substantially above the main full-band Block 1 result. That figure should nevertheless be interpreted with caution, since the exported arrays still show Block 1 labels only. For that reason, the appendix documents this auxiliary Block 1+Block 2 analysis while the main empirical interpretation remains anchored to the clearly defined Block 1 export.

Chapter 7

Discussion

7.1 Data cleaning as part of the scientific result

One of the clearest lessons of this study is that cleaning should not be treated as a backstage technicality. The decisive removal rules were not arbitrary. The final exclusion set was dominated by two reasons, both directly tied to the main artifact concern: stimulus interruption by DC and stimulus onset within the estimated post-DC recovery period. This matters scientifically because it reduces the risk that the classifier is learning artifact-related transients instead of cue-related anticipatory activity.

The thesis also required one important correction relative to the older draft: the final Block 1 training analysis did not use the block-specific 1.827 s DC tail described there. Instead, it applied 1.836 s, the value adopted in the final classification study. This is exactly the kind of mismatch that matters in a methods chapter, because a thesis must describe the threshold that actually governed the final dataset, not merely a nearby value from an earlier recovery analysis.

7.2 Interpreting the full-band model comparison

The main full-band comparison suggests that an explicit time–frequency representation was not, by itself, the best route to subject-generalizing performance in this dataset. The spectrogram CNN achieved the strongest validation performance but the weakest held-out generalization among the three main full-band models. Several explanations are plausible.

First, the STFT representation compresses the cue period into only eight time bins. That may be sufficient for broad oscillatory summaries, but it may also blur temporal details that matter when the task is to distinguish subtle anticipatory states across unseen individuals. Second, the CNN may have exploited spectral regularities that were stable enough to support validation selection yet not robust enough to transfer to new subjects. Third, the fixed STFT configuration was retained for

coherence and comparability, not exhaustively tuned. The present findings therefore should not be read as a general failure of time–frequency methods. Rather, they indicate that under the retained preprocessing and search ranges, the strongest subject-generalizing representation was the one that preserved more of the original cue-period temporal structure.

The relative stability of the attention TCN is theoretically interesting. Compared with the baseline TCN, the attention-augmented variant adds only a modest increase in model capacity, yet it produced the best held-out result and the smallest validation-to-test degradation. That pattern suggests that the main advantage was not simply model size. A more plausible interpretation is that adaptive temporal weighting helped the network focus on the most informative portions of the cue interval while remaining less vulnerable than the spectrogram CNN to validation-specific regularities. In a problem where the signal is weak, subject variability is substantial, and the target is cognitively rather than sensorily defined, such temporal summarization may be a better inductive bias than aggressive compression into a small number of time bins.

7.3 Alpha, theta, and bodily specificity

The exploratory variants align well with the qualitative background provided by both the prior literature and the project context. The alpha-only attention TCN produced the strongest held-out result across all analysed settings. This is the pattern one would expect if somatomotor alpha carries especially informative body-specific anticipatory information [7, 8, 9, 37, 4, 40, 41]. The result does not prove that alpha is the only relevant source, but it does suggest that restricting the input to the alpha range can improve the signal-to-noise ratio for the specific left-versus-right-plus-safety decoding problem. In theoretical terms, that interpretation fits both the broader pain-anticipation literature and accounts of alpha as a mechanism that shapes sensorimotor readiness through selective gating.

The theta-only variants still performed above or near chance, but they did not reach the level of the best alpha-only model. This is theoretically coherent [32, 4, 5, 24]. A more general threat-related or control-related process can still support the present classification problem, especially for distinguishing threatening cues from safety, but it is not necessarily the most spatially selective feature source. In other words, theta may contribute to the decoding problem without being the main carrier of laterality. That interpretation also fits the broader project account, in which theta is linked more strongly to threat expression during acquisition and to updating-related

demands during reversal than to a sharply lateralized bodily code [40, 41].

The C3/C4-only results provide a useful caution. It would be tempting to assume that if the signal is somatomotor, then only the two canonical central channels should matter. The exploratory analysis argues against that simplification. Restricting the input too strongly weakened performance. This suggests that although bodily specificity is relevant, the decodable pattern is distributed enough that the models benefit from channels beyond the most obvious sensorimotor pair. In other words, the bodily signal appears to be concentrated enough for alpha-based decoding to help, yet broad enough that it is not reducible to a narrowly local two-channel representation.

7.4 Relation to the broader multimodal project

The broader project provides a useful external frame for interpreting the EEG-only decoding results. In the internal experiment illustration and poster, somatomotor alpha and corticospinal excitability are both described as carrying information about threat value and threatened body side, whereas SCR is described as a non-lateralized index of autonomic threat responding [40, 41]. The present decoding pattern fits that contrast reasonably well. The strongest exploratory result was obtained in the alpha-restricted setting, which is consistent with the idea that laterality-related information is especially concentrated in alpha-linked preparatory dynamics. By contrast, theta remained informative but did not dominate, which fits the project-level interpretation of theta as a broader threat and updating signal rather than the most side-specific code.

The MEP findings are particularly relevant here. The experiment illustration reports limb-specific MEP inhibition for threat cues and highlights that alpha, but not theta, predicted MEP changes [40]. Related work on pain expectancy and corticospinal excitability points in the same direction, showing that anticipation of pain can recruit motor inhibition before pain occurs and that these effects can be topographically organized [33, 34]. The present thesis does not decode MEPs and does not establish a direct causal relation between EEG features and corticospinal output. Nevertheless, the alignment between the alpha-only decoding advantage and the broader alpha–MEP project account strengthens the interpretation that the decoded information is not merely generic arousal.

SCR provides the complementary side of the story. The project materials describe SCR increases for threat cues regardless of expected body side [40, 41]. That pattern is useful because it shows that not every modality in the broader experiment is

expected to carry spatial information in the same way. The fact that the present thesis still finds some informative signal in theta- and full-band settings, but the clearest advantage in alpha, is therefore plausible in multimodal terms: the anticipatory state likely includes both a body-specific component and a broader non-lateralized threat component. The current EEG results suggest that the former is recoverable, but only modestly, and under conditions where careful cleaning and conservative evaluation are maintained.

7.5 Strengths of the present study

Several strengths stand out in the implementation and the thesis.

Traceability. The pipeline preserves enough metadata to map samples back to original recordings and annotation contexts.

Targeted cleaning. The main removals are interpretable and tied to clearly articulated contamination mechanisms.

Leakage control. Grouped subject-wise splitting, train-only normalization, and validation-only model selection directly address common EEG evaluation pitfalls.

Comparative framing. The thesis does not make its claim depend on a single architecture. It compares multiple families and exploratory restrictions while keeping the cleaned source dataset fixed.

7.6 Limitations

The thesis also has clear limitations, and stating them explicitly helps keep the interpretation proportionate.

Single grouped holdout split. The reported numbers come from one grouped train/validation/test partition rather than repeated grouped cross-validation or repeated grouped holdout.

No permutation-based decoding significance. The results include accuracy and macro-F1, but not permutation-based significance testing or confidence intervals.

Modest subject count. Although 1711 clean epochs is a useful sample size, they come from 29 exported subjects/recordings. Subject-wise EEG generalization remains difficult at that scale.

No completed end-to-end no-tail ablation. The dataset-audit stage explicitly noted that the paired comparison datasets needed for a no-tail versus with-tail downstream ablation were not present. The thesis can therefore justify the DC-tail

rule strongly, but it cannot claim a completed no-tail versus with-tail decoding comparison.

Classical ablation code present but not executed. The dataset-audit analysis also contains a grouped cross-validation ablation path based on `StandardScaler`, fold-wise `PCA`, and `LogisticRegression` with the `saga` solver. This path was designed to compare "no DC tail" and "with DC tail" NPZ exports in either three-class or left-versus-right mode, but the required NPZ files were not present, so no classical baseline result could be reported.

Raw-scan versus export discrepancy. The 30-to-29 discrepancy between scanned raw files and exported recording identifiers is verified, but its cause cannot be resolved conclusively from the retained summaries.

Block 1 only. Restricting the analysis to acquisition is scientifically motivated, but it means that the present results do not establish how the same pipeline would behave during reversal.

7.7 Future work

The present study suggests several natural next steps. First, the same cleaning and split logic should be extended to Block 2 reversal so that stable anticipation and updating-related anticipation can be compared under a matched protocol. Second, repeated grouped resampling or grouped cross-validation with confidence intervals would help quantify uncertainty around the decoding results. Third, a completed no-tail versus with-tail comparison would make the downstream effect of contamination control even more explicit. Fourth, alpha-focused time-domain models deserve closer study, for example through spatial attention, source-informed channel grouping, or more interpretable saliency analyses. Fifth, multimodal fusion with MEP or SCR information could test whether spatial anticipation is decoded best from EEG alone or from integrated central-plus-peripheral representations.

Chapter 8

Conclusion

This thesis asked whether cue-period EEG in Block 1 of a Pavlovian threat-learning experiment contains enough information to decode the expected spatial content of pain, distinguishing left-threat, right-threat, and safety cues. On the basis of the evidence assembled in this work, the answer is yes, provided that it is stated with appropriate caution.

The strongest contribution of the work is not simply the final classification score. It lies in the combination of a traceable segmentation pipeline, data-driven justification of the cue-duration tolerance, explicit handling of DC-related contamination, and a leakage-free grouped subject-wise evaluation protocol. Within that framework, the best full-band model — an attention-augmented TCN — reached 39.93% test accuracy and macro-F1 0.3867. The exploratory analyses further suggested that alpha-restricted time-domain input is especially informative, with the alpha-only attention TCN reaching 40.27% and macro-F1 0.3975. That pattern is consistent with a broader literature linking somatomotor alpha to anticipatory sensorimotor regulation and with project-level findings relating alpha, but not theta, to corticospinal excitability [7, 8, 9, 40, 41].

The results remain moderate, and the thesis does not overstate them. They do not show that the task is easy, nor that any single architecture solves it. They show that spatial anticipation is recoverable, but only modestly, from cleaned EEG recorded in an artifact-prone multimodal setting. This is precisely why the methodological argument matters: without justified cleaning and evaluation, the positive result would be less trustworthy; with them, even a moderate result becomes scientifically useful.

Appendix A

Reproducibility Notes and Data Dictionary

A.1 Shapes of the exported arrays

The main exported arrays had the shapes reported in Table A.1. The same split sizes were used across the full-band, alpha-only, theta-only, and C3/C4-only variants.

Table A.1: Main tensor shapes reported by the final analyses.

Setting	Time-domain shape	STFT shape	Notes
Full-band Block 1	$(1711, 64, 1125)$	$(1711, 64, 481, 8)$	Main confirmatory run
Alpha-only	$(1711, 64, 1125)$	$(1711, 64, 481, 8)$	8–12 Hz band-pass before export
Theta-only	$(1711, 64, 1125)$	$(1711, 64, 481, 8)$	4–8 Hz band-pass before export
C3/C4-only	$(1711, 2, 1125)$	$(1711, 2, 481, 8)$	Channel restriction only

A.2 Event and segment categories encountered in the raw scan

The main Block 1 analysis reported the global segment counts shown in Table A.2 across the scanned recordings before cleaning. Presenting these counts as a table rather than a long bullet list makes the symmetry of the annotation stream easier to see.

Table A.2: Global segment counts reported by the final main analysis before cleaning.

Segment category	Count
New recording segment	30
DC-correction segment	144
No-stimulus cue segment	580
TMS-after-no-stimulus segment	580
Left-stimulus cue (reinforced)	406
TMS after left-stimulus cue (reinforced)	406
Right-stimulus cue (reinforced)	406
TMS after right-stimulus cue (reinforced)	406
Right-stimulus cue (non-reinforced)	174
TMS after right-stimulus cue (non-reinforced)	174
Left-stimulus cue (non-reinforced)	174
TMS after left-stimulus cue (non-reinforced)	174

These counts confirm that the candidate task space is symmetric at the annotation level: before cleaning, the stream contains 580 left-cue instances, 580 right-cue instances, and 580 safety-cue instances.

A.3 Per-recording removal profile from the dataset-audit analysis

The dataset-audit analysis provides a compact per-recording removal summary. In the visible printed rows, most recordings retained all 60 candidate task epochs, whereas a small number accounted for the removals observed in the final audit. The most affected printed example was `03as\_block1` with 55 kept out of 60 candidate epochs, corresponding to 5 removed epochs (8.33%). The printed rows for `02gd\_block1`, `09dm\_block1`, and `10lc\_block1` each showed 1 removed epoch out of 60 (1.67%). This supports the broader interpretation that the cleaning stage was targeted rather than globally punitive.

A.4 Audit note on raw-scan versus exported dataset counts

Two verified counts should be kept conceptually distinct. First, the Block 1 raw-data scan covered 30 recording files. Second, the final machine-learning export used for

splitting and model training contains 29 unique recording identifiers and 1711 clean task epochs. These quantities refer to different stages of the study and should not be treated as interchangeable.

Table A.3: Compact audit note distinguishing the Block 1 raw scan from the exported dataset used for learning.

Pipeline stage	Verified count
Raw Block 1 files scanned	30
Exported unique recording identifiers	29
Retained clean task epochs in final machine-learning dataset	1711
Exact cause of the 30-to-29 discrepancy	Not fully resolvable from the available summaries
Scope of the final empirical claims	The exported dataset used for learning

Appendix B

Hyperparameter Search Grids

This appendix reports the bounded search grids used in the final analyses.

B.1 Time-domain baseline grid

The full baseline TCN grid comprised the Cartesian product of the following values:

- channels in $\{(32, 64, 96), (64, 96, 128)\}$;
- kernels in $\{(9, 7, 5), (11, 9, 7)\}$;
- dropout in $\{0.20, 0.30\}$;
- learning rate in $\{10^{-4}, 3 \times 10^{-4}\}$;
- weight decay in $\{10^{-4}, 2 \times 10^{-4}\}$;
- label smoothing $= 0.05$;
- batch size $= 8$.

This yields 32 configurations, of which 12 were sampled in the study.

B.2 Time-domain attention grid

The full attention-TCN grid added one more factor:

- attention hidden size in $\{16, 32\}$.

All other factors matched the baseline grid, yielding 64 configurations in total, of which 16 were evaluated in each analysis.

B.3 Spectrogram grid

The full spectrogram CNN grid comprised:

- base channels in $\{16, 32\}$;
- dropout in $\{0.20, 0.30\}$;
- learning rate in $\{10^{-4}, 3 \times 10^{-4}\}$;
- weight decay in $\{10^{-4}, 10^{-3}\}$;
- batch size in $\{16, 32\}$;
- label smoothing $= 0.05$.

This yields 32 configurations, of which 16 were evaluated in each analysis.

Appendix C

Selected Configurations and Supplementary Result Tables

C.1 Best configurations reported by the final analyses

Table C.1 summarizes the selected configurations across the main and exploratory settings.

Table C.1: Best configurations reported by the final analyses.

Setting	Model family	Configuration
Full-band	Baseline TCN	channels=(64,96,128), kernels=(9,7,5), dropout=0.2, lr=1e-4, wd=2e-4, batch=8
Full-band	Attention TCN	channels=(64,96,128), kernels=(11,9,7), dropout=0.2, att_hidden=32, lr=1e-4, wd=1e-4, batch=8
Full-band	Spectrogram CNN	base_channels=32, dropout=0.3, lr=3e-4, wd=1e-4, batch=32
Alpha-only	Baseline TCN	channels=(32,64,96), kernels=(11,9,7), dropout=0.3, lr=3e-4, wd=2e-4, batch=8
Alpha-only	Attention TCN	channels=(32,64,96), kernels=(9,7,5), dropout=0.2, att_hidden=16, lr=3e-4, wd=2e-4, batch=8
Alpha-only	Spectrogram CNN	base_channels=32, dropout=0.3, lr=3e-4, wd=1e-4, batch=32
Theta-only	Baseline TCN	channels=(32,64,96), kernels=(9,7,5), dropout=0.3, lr=3e-4, wd=1e-4, batch=8
Theta-only	Attention TCN	channels=(64,96,128), kernels=(11,9,7), dropout=0.3, att_hidden=32, lr=1e-4, wd=2e-4, batch=8

Table C.1: Best configurations reported by the final analyses (continued).

Setting	Model family	Configuration
Theta-only	Spectrogram CNN	base_channels=32, dropout=0.2, lr=3e-4, wd=1e-4, batch=16
C3/C4-only	Baseline TCN	channels=(64,96,128), kernels=(9,7,5), dropout=0.2, lr=1e-4, wd=1e-4, batch=8
C3/C4-only	Attention TCN	channels=(64,96,128), kernels=(11,9,7), dropout=0.3, att_hidden=16, lr=1e-4, wd=2e-4, batch=8
C3/C4-only	Spectrogram CNN	base_channels=16, dropout=0.3, lr=1e-4, wd=1e-3, batch=16

Appendix D

Supplementary Diagnostic Figures

The modelling analyses provide more than the scalar scores reported in the main chapters. Each setting also provides compact validation-versus-test summaries, train/validation curves, learning-rate schedules, and final confusion matrices. This appendix gathers those supplementary diagnostics so that the numerical comparisons in the main text can be read alongside the optimisation behaviour and class-specific error patterns of the selected models.

D.1 Alpha-restricted analysis

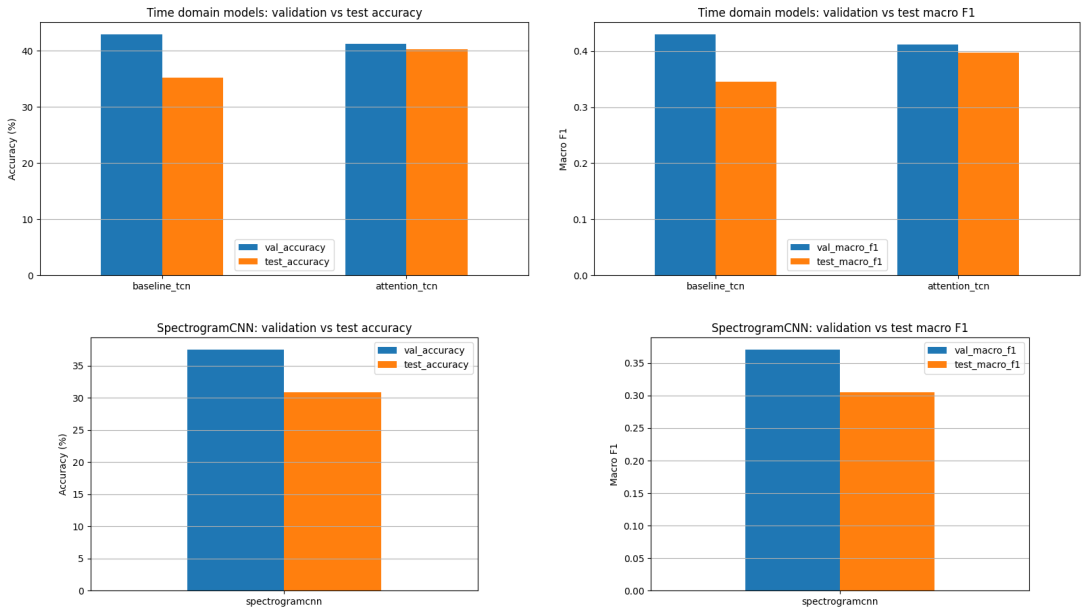

Figure D.1: Summary figures for the alpha-restricted analysis. The grid presents the compact held-out validation-versus-test comparisons for both the time-domain and spectrogram analyses.

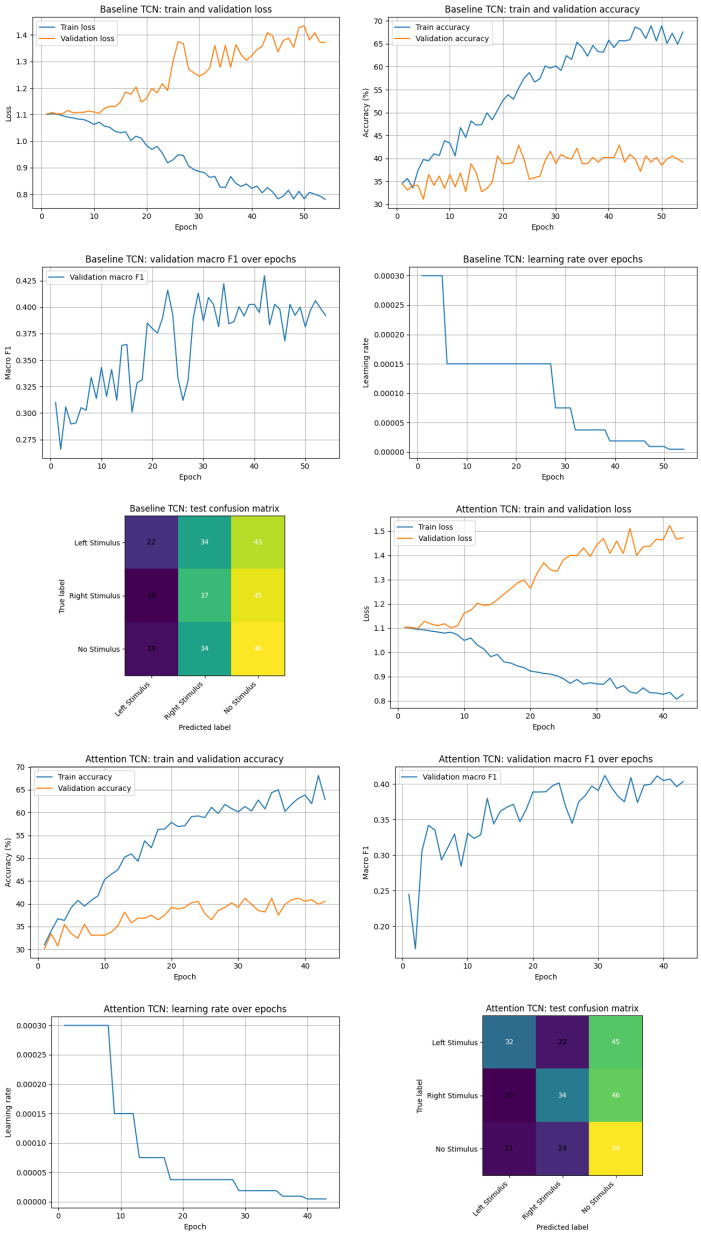

Figure D.2: Time-domain diagnostic figures for the alpha-restricted analysis, including the learning curves, learning-rate schedules, and confusion matrices of the selected baseline and attention TCN runs.

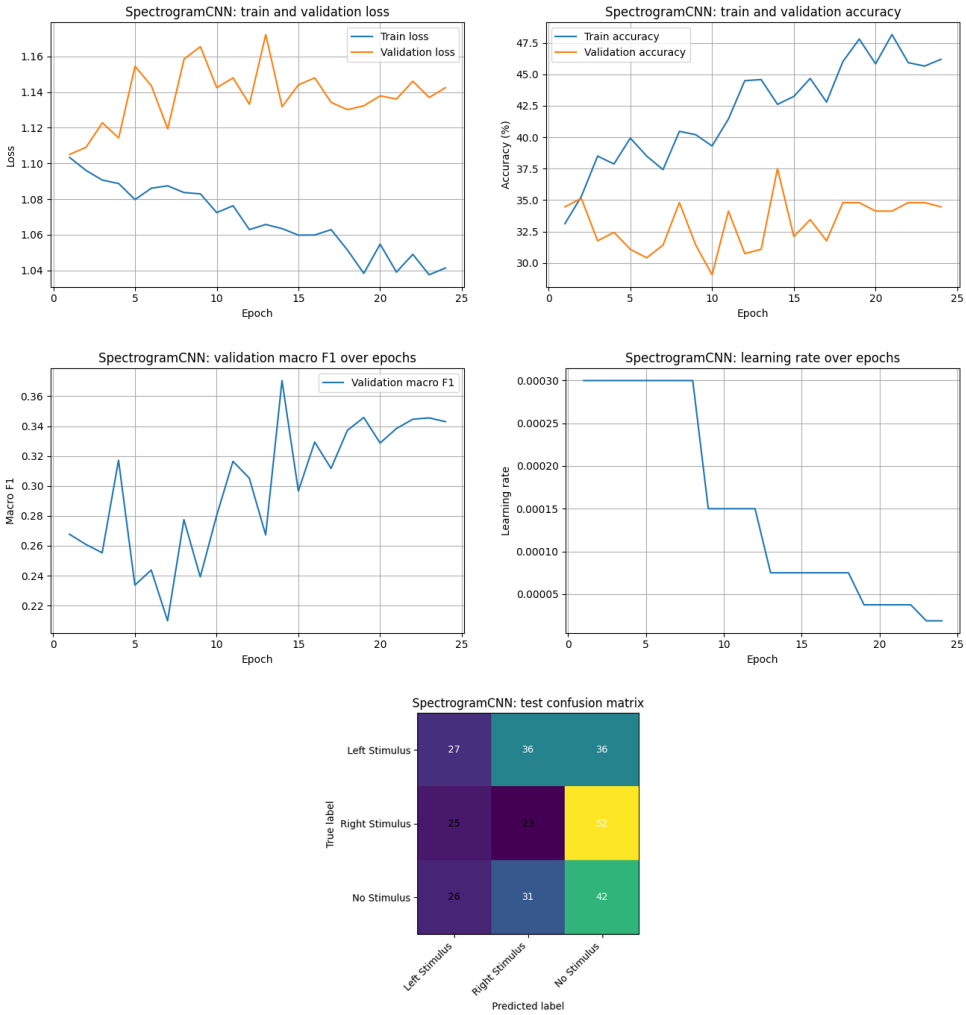

Figure D.3: Time-frequency diagnostic figures for the alpha-restricted analysis, including the learning curves, learning-rate schedule, and confusion matrix of the selected spectrogram CNN.

D.2 Theta-restricted analysis

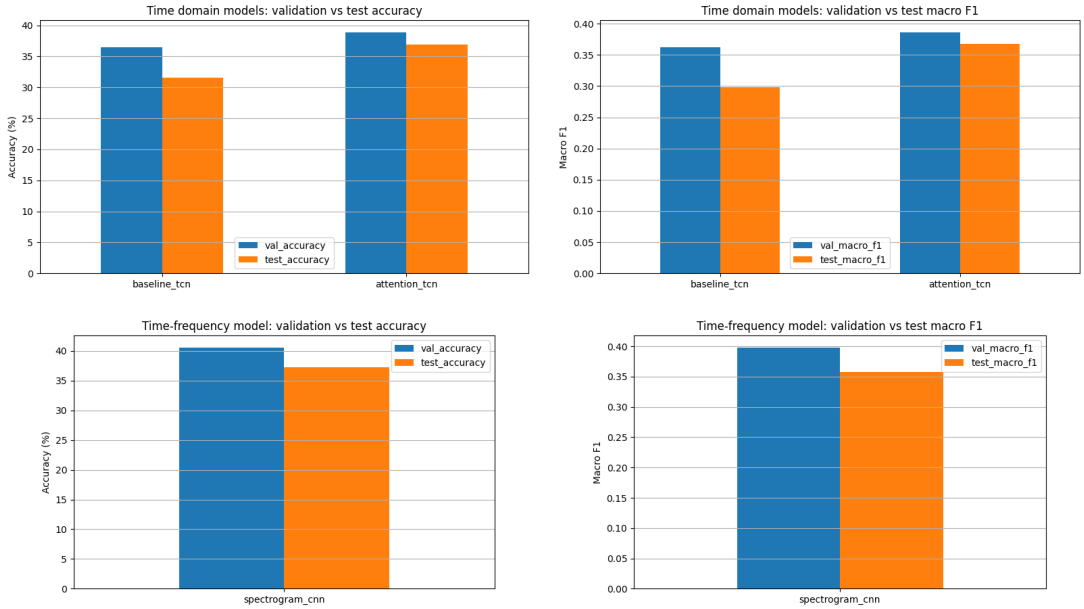

Figure D.4: Summary figures for the theta-restricted analysis. The grid shows the compact held-out summaries for the theta-restricted analyses in both representations.

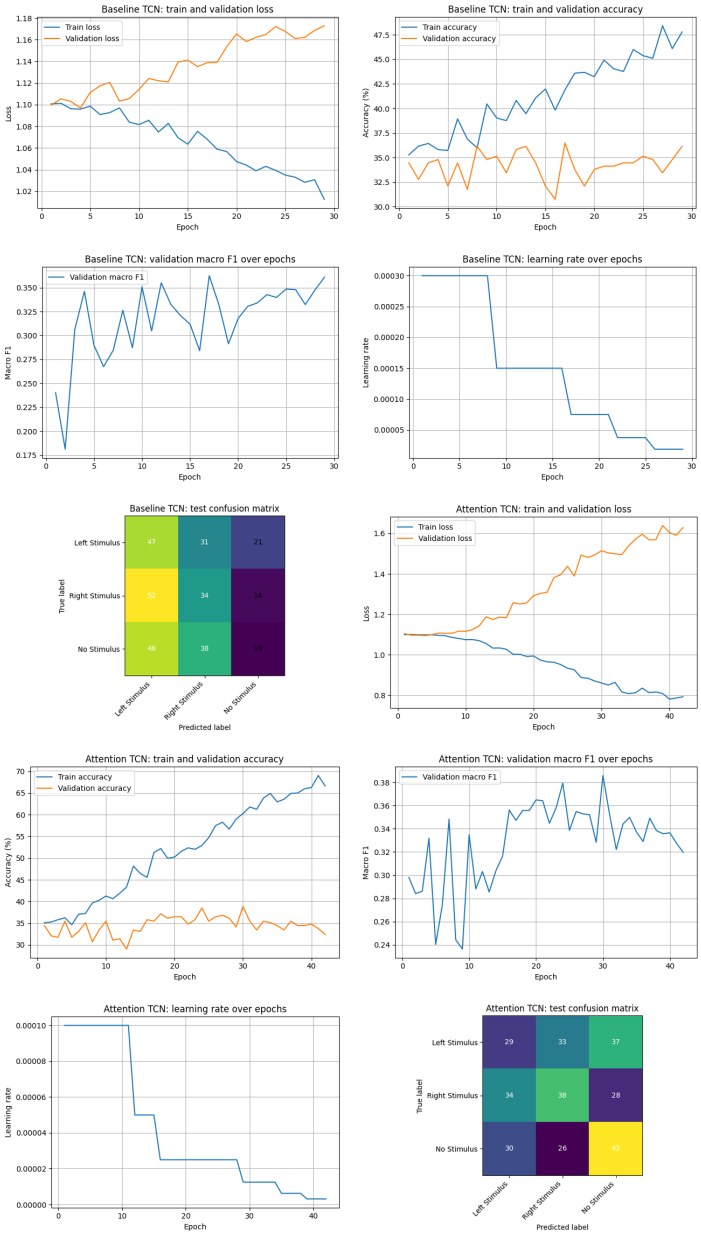

Figure D.5: Time-domain diagnostic figures for the theta-restricted analysis, including the learning curves, learning-rate schedules, and confusion matrices of the selected baseline and attention TCN runs.

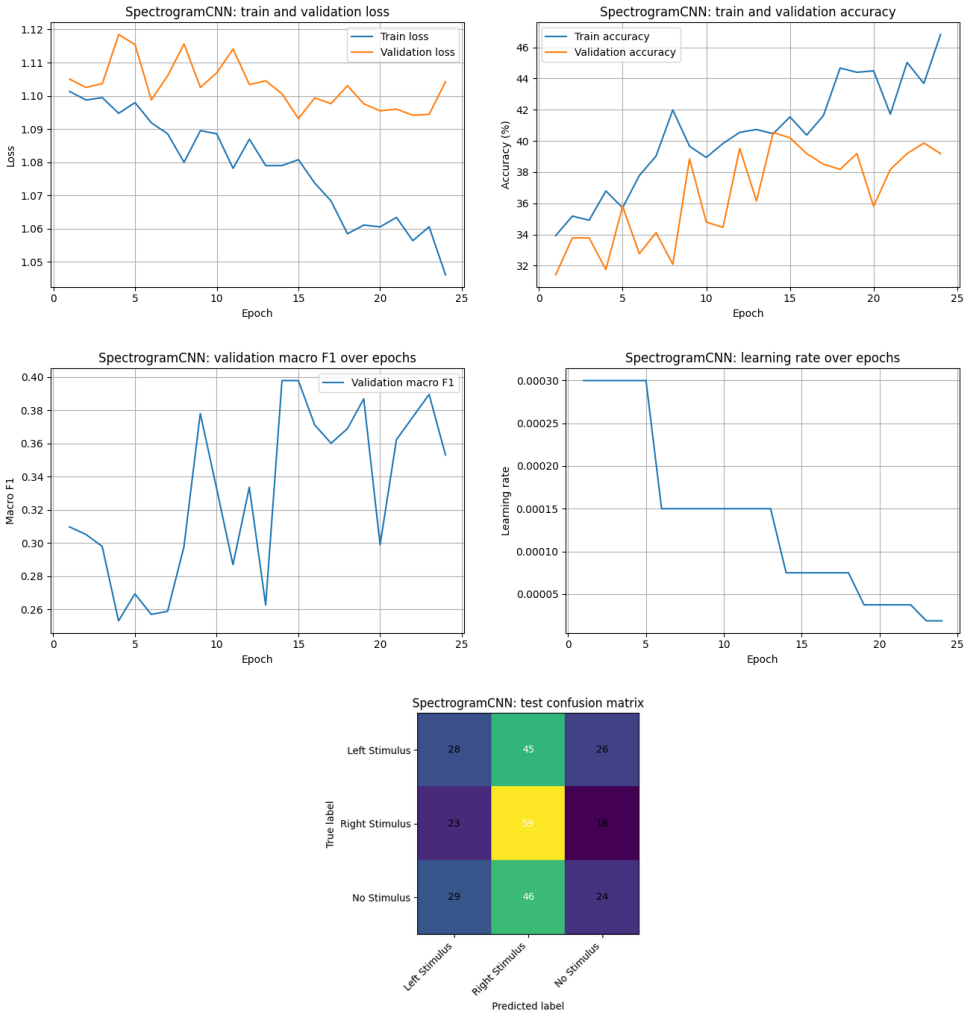

Figure D.6: Time-frequency diagnostic figures for the theta-restricted analysis, including the learning curves, learning-rate schedule, and confusion matrix of the selected spectrogram CNN.

D.3 C3/C4-restricted analysis

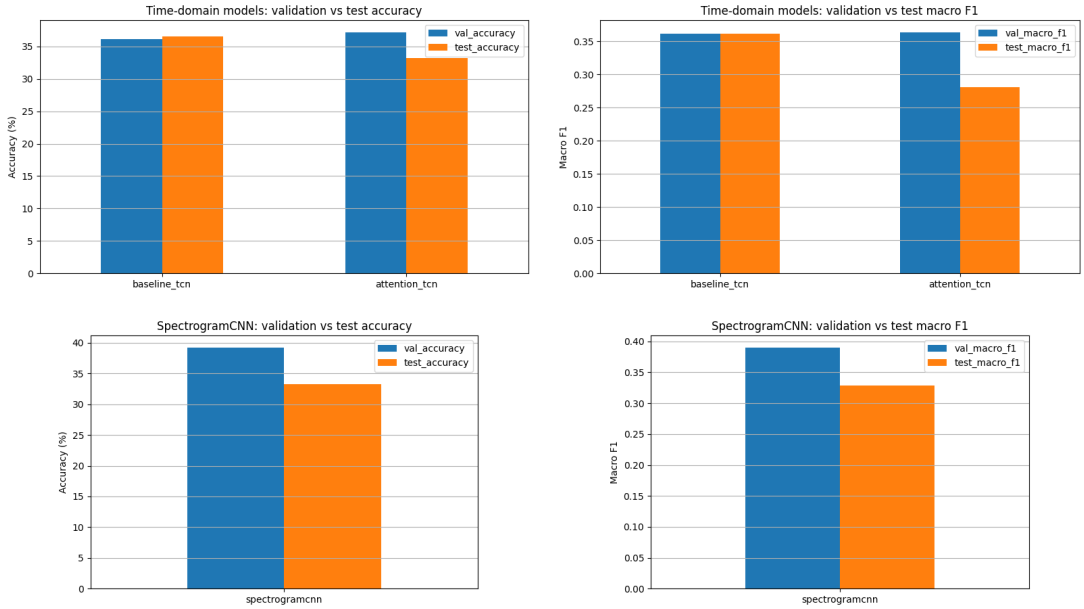

Figure D.7: Summary figures for the C3/C4-restricted analysis. The reduced-input results are shown here in the same reporting format used for the other full-channel analyses.

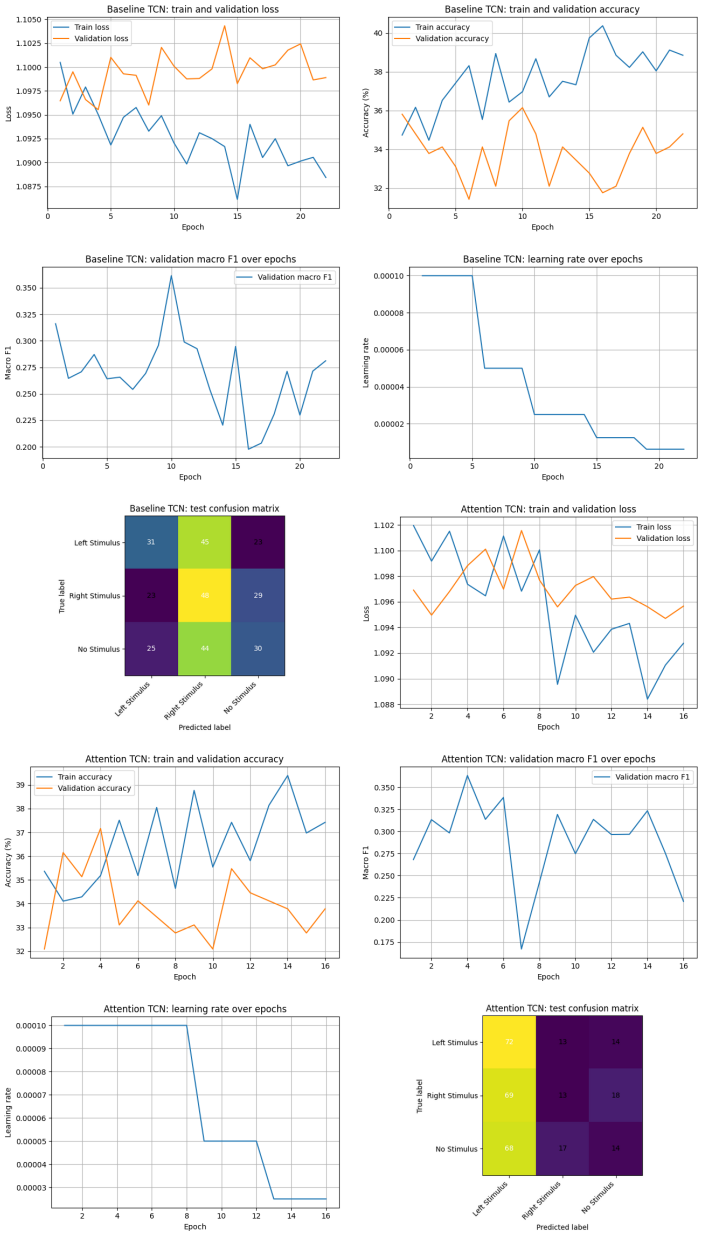

Figure D.8: Time-domain diagnostic figures for the C3/C4-restricted analysis, including the learning curves, learning-rate schedules, and confusion matrices of the selected baseline and attention TCN runs.

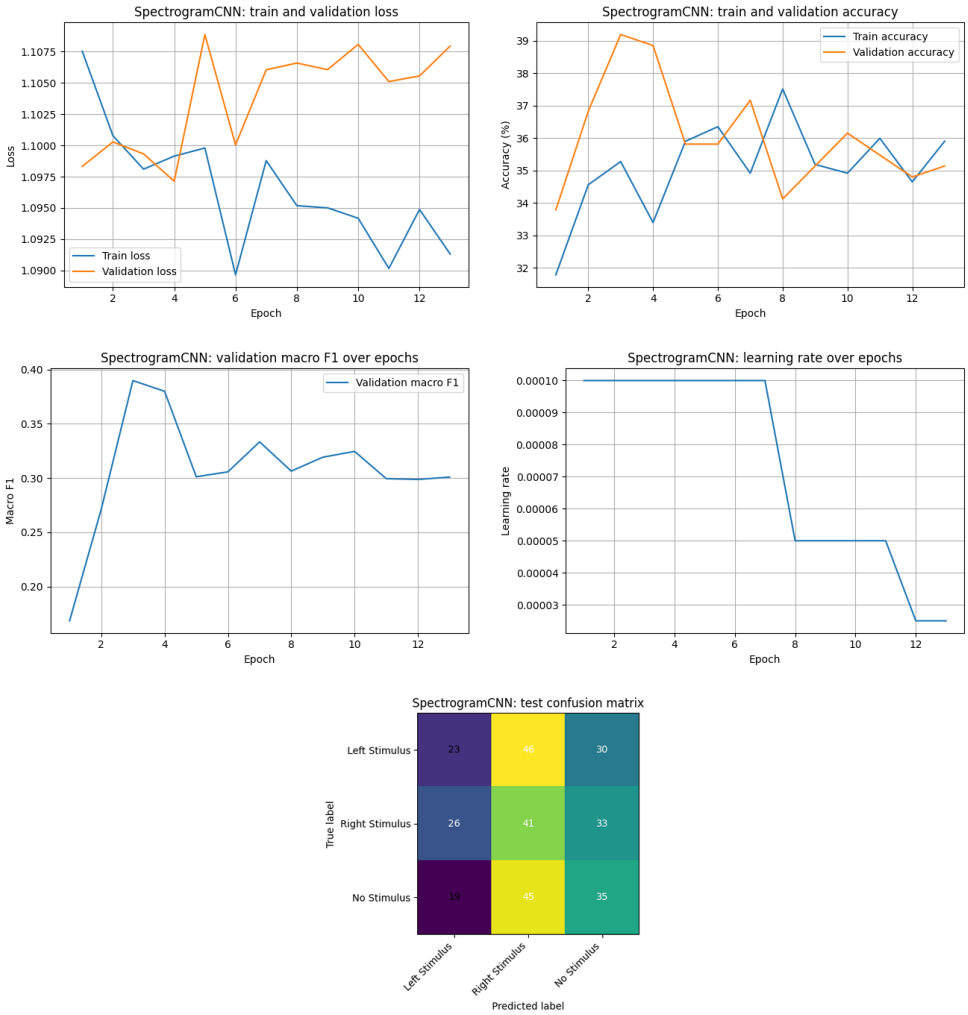

Figure D.9: Time-frequency diagnostic figures for the C3/C4-restricted analysis, including the learning curves, learning-rate schedule, and confusion matrix of the selected spectrogram CNN.

D.4 Auxiliary Block 1+Block 2 analysis

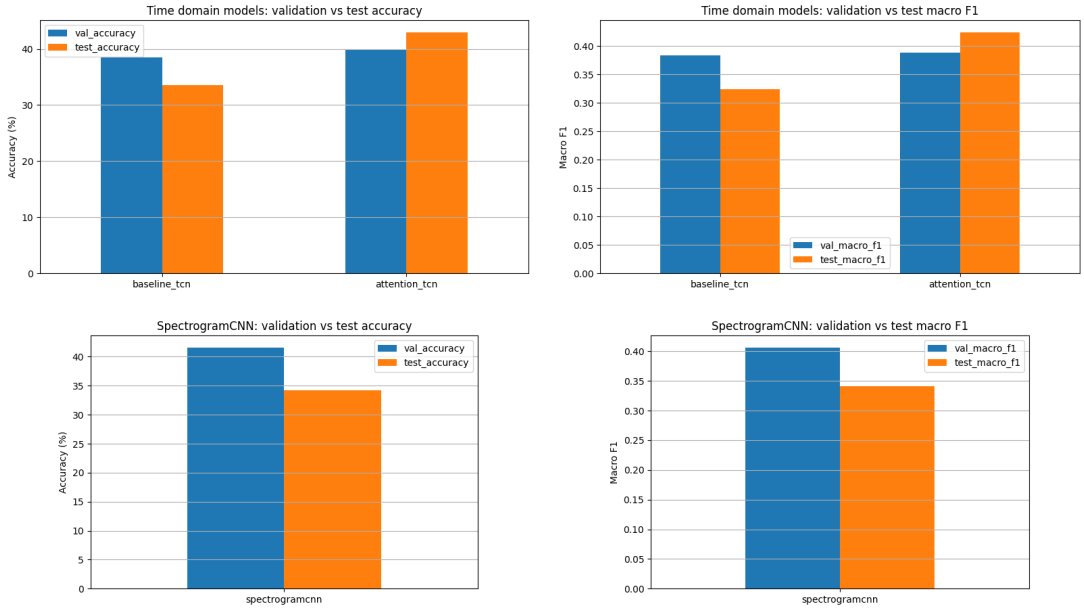

Figure D.10: Summary figures from the auxiliary Block 1+Block 2 analysis. They are included as supplementary material to document the auxiliary Block 1+Block 2 comparison mentioned in the main text.

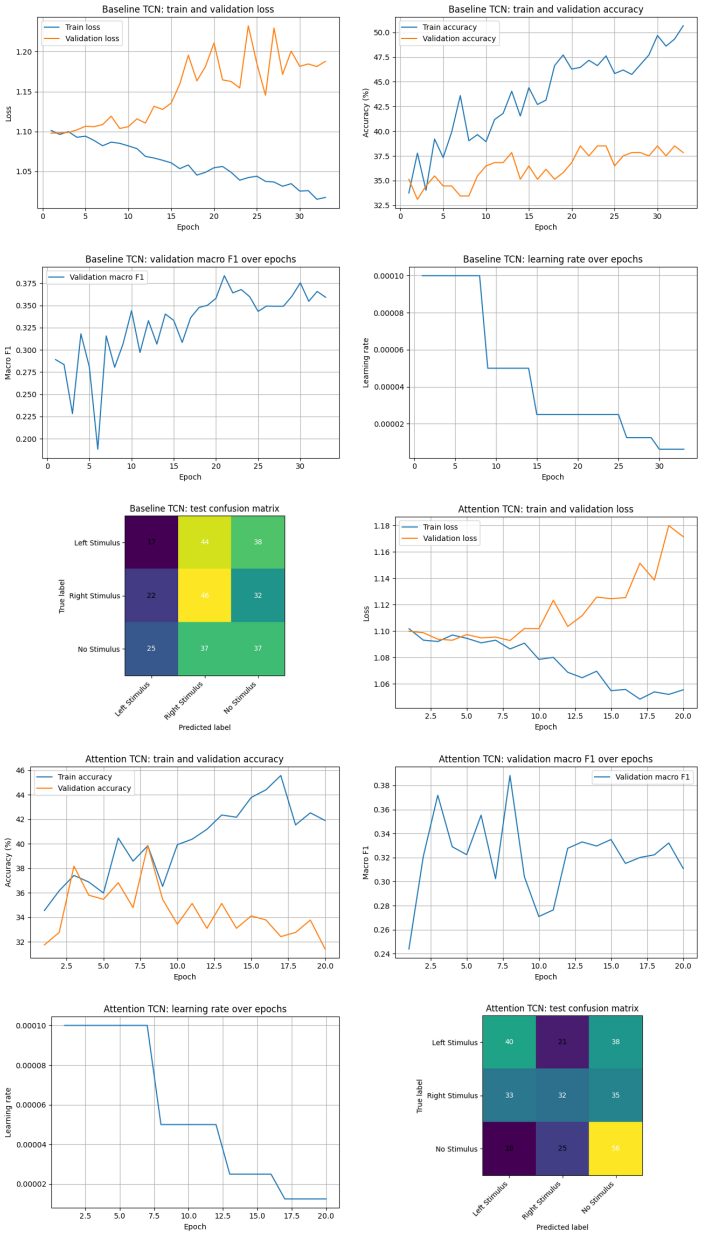

Figure D.11: Time-domain diagnostic figures from the auxiliary Block 1+Block 2 analysis. They are included as supplementary material for the auxiliary Block 1+Block 2 comparison.

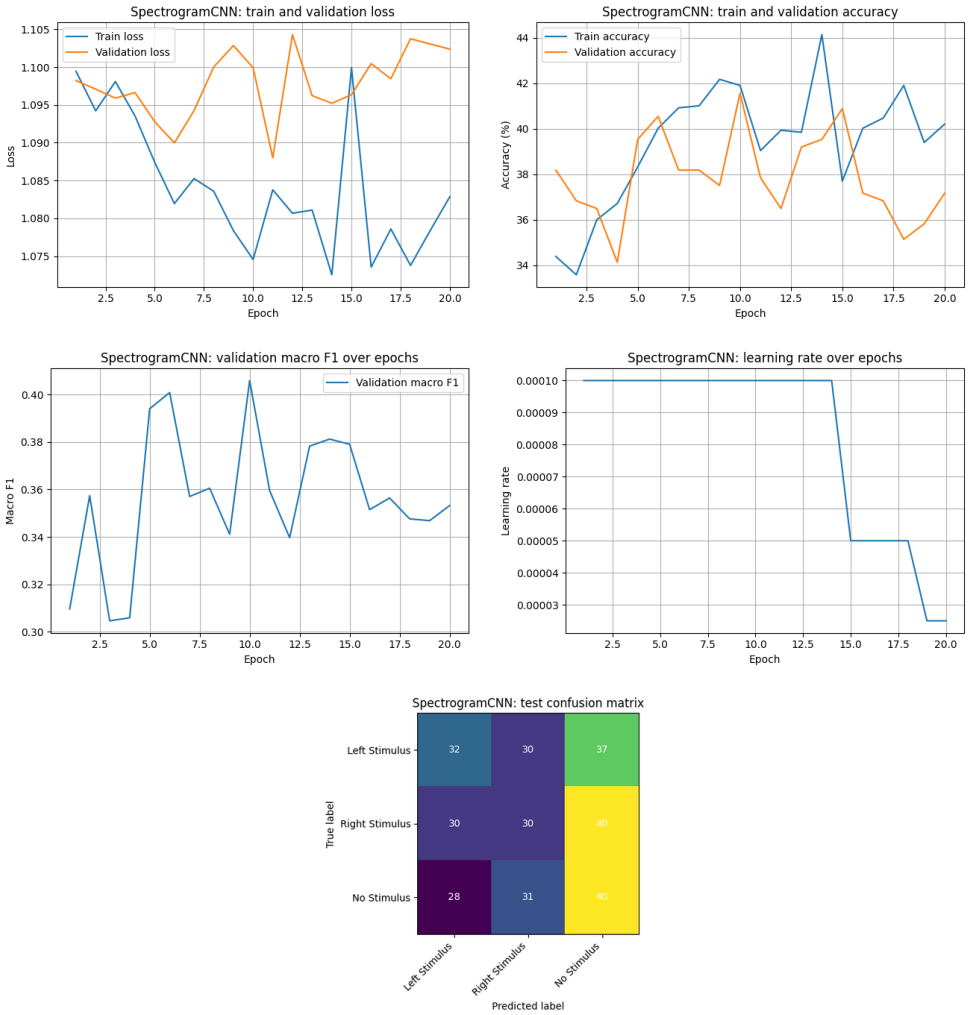

Figure D.12: Time-frequency diagnostic figures from the auxiliary Block 1+Block 2 analysis. The grid shows the selected spectrogram diagnostics for the auxiliary Block 1+Block 2 analysis.

D.5 Supplementary best-result tables

Table D.1 and Table D.2 report the best result for each model family in each analytical setting.

Table D.1: Best results for the main full-band, alpha-only, and theta-only settings.

Analysis setting	Model family	Params	Val. macro-F1	Val. acc. (%)	Test macro-F1	Test acc. (%)
Main full-band	Baseline TCN	150755	0.3950	39.53	0.3637	36.58
Main full-band	Attention TCN	199972	0.3839	38.85	0.3867	39.93
Main full-band	Spectrogram CNN	111651	0.4253	42.57	0.3535	35.23
Alpha-only	Baseline TCN	89475	0.4297	42.91	0.3448	35.23
Alpha-only	Attention TCN	70564	0.4121	41.22	0.3975	40.27
Alpha-only	Spectrogram CNN	111651	0.3705	37.50	0.3047	30.87
Theta-only	Baseline TCN	68995	0.3625	36.49	0.2980	31.54
Theta-only	Attention TCN	199972	0.3859	38.85	0.3678	36.91
Theta-only	Spectrogram CNN	111651	0.3978	40.54	0.3574	37.25

Table D.2: Best results for the C3/C4-only setting and the auxiliary Block 1+Block 2 analysis.

Analysis setting	Model family	Params	Val. macro-F1	Val. acc. (%)	Test macro-F1	Test acc. (%)
C3/C4-only	Baseline TCN	114919	0.3615	36.15	0.3616	36.58
C3/C4-only	Attention TCN	154120	0.3632	37.16	0.2808	33.22
C3/C4-only	Spectrogram CNN	23859	0.3897	39.19	0.3281	33.22
Auxiliary Block 1+Block 2	Baseline TCN	150755	0.3834	38.51	0.3244	33.56
Auxiliary Block 1+Block 2	Attention TCN	91044	0.3882	39.86	0.4239	42.95
Auxiliary Block 1+Block 2	Spectrogram CNN	32787	0.4059	41.55	0.3410	34.23

Appendix E

Supplementary Analytical Tables and Notes

E.1 Comparative overview of analytical variants

The analytical variants are easier to compare when split into two complementary tables rather than forced into one oversized matrix. Table E.1 records the data scope and exported tensor shapes. Table E.2 records the strongest held-out result in each model family.

Table E.1: Comparative overview of data scope and exported representations across analytical variants.

Analysis setting	Run label or scope	Band-pass	Retained channels	Exported time-domain shape	Exported STFT shape
Main full-band	Block 1 confirmatory run	0.5–100 Hz	All EEG channels	$(1711, 64, 1125)$	$(1711, 64, 481, 8)$
Alpha-only	Block 1 alpha-restricted run	8–12 Hz	All EEG channels	$(1711, 64, 1125)$	$(1711, 64, 481, 8)$
Theta-only	Block 1 theta-restricted run	4–8 Hz	All EEG channels	$(1711, 64, 1125)$	$(1711, 64, 481, 8)$
C3/C4-only	Block 1 channel-restricted run	0.5–100 Hz	C3 and C4 only	$(1711, 2, 1125)$	$(1711, 2, 481, 8)$
Auxiliary Block 1+Block 2	Run labelled Block 1+Block 2	0.5–100 Hz	All EEG channels	$(1711, 64, 1125)$	$(1711, 64, 481, 8)$

Table E.2: Strongest held-out result in each model family. Accuracy is reported first, followed by macro-F1.

Analysis setting	Strongest held-out result(s)
Main full-band	Attention TCN: 39.93% / 0.3867; baseline TCN: 36.58% / 0.3637; spectrogram CNN: 35.23% / 0.3535
Alpha-only	Attention TCN: 40.27% / 0.3975; baseline TCN: 35.23% / 0.3448; spectrogram CNN: 30.87% / 0.3047
Theta-only	Spectrogram CNN: 37.25% / 0.3574; attention TCN: 36.91% / 0.3678; baseline TCN: 31.54% / 0.2980
C3/C4-only	Baseline TCN: 36.58% / 0.3616; attention TCN: 33.22% / 0.2808; spectrogram CNN: 33.22% / 0.3281
Auxiliary Block 1+Block 2	Attention TCN: 42.95% / 0.4239; baseline TCN: 33.56% / 0.3244; spectrogram CNN: 34.23% / 0.3410

E.2 Interpretive notes on analytical variants

Table E.1 and Table E.2 underscore two points that are useful for interpretation. First, the exploratory analyses are not filtered copies of a single result table: each setting has its own hyperparameter search, learning curves, and confusion matrices. In particular, the theta-restricted setting includes complete baseline and attention TCN searches in the time domain as well as the spectrogram model. Second, the auxiliary Block 1+Block 2 analysis should be read cautiously. Its initial scan covers both block folders and its overall label counts differ from those of the Block 1 analyses, yet the later split summary still reports only Block 1 labels in the arrays used for learning. For that reason, it is included here as supplementary material rather than as part of the main empirical basis of the thesis.

A smaller internal difference concerns the fine-label naming of TMS-adjacent segments. The main Block 1 analyses use labels such as `tms\_left\_real`, `tms\_right\_real`, and `tms\_no`, whereas the auxiliary 2-block analysis uses labels such as `tms\_after\_left`, `tms\_after\_right`, and `tms\_after\_no`, and also reports a large number of `unknown` segments in its overall label counts. These differences do not affect the final three-class cue labels used for learning, but they clarify how the Block 1+Block 2-labelled auxiliary analysis was annotated internally.

E.3 Auxiliary methodological components

Four auxiliary methodological components deserve brief mention because they support key parts of the thesis even though they are not themselves the principal modelling analysis.

Table E.3: Auxiliary methodological components and their role in the thesis.

Auxiliary component	Status in the final thesis	Contribution to the thesis
Block 1 DC-recovery analysis	Auxiliary but load-bearing	Provides the earlier block-specific recovery statistics and the initial 1.827 s post-DC recommendation
Combined-block DC-recovery analysis	Auxiliary but load-bearing	Provides the combined-block recovery scan and the later shared 1.836 s recommendation that the final modelling analyses actually applied
Block 1 dataset-audit analysis	Auxiliary but load-bearing	Provides the retained removal counts, per-recording dataset summaries, the stimulus-tolerance estimate, STFT-justification outputs, and the documented but non-executed PCA plus logistic-regression ablation path
Combined-block dataset-audit analysis	Auxiliary consistency analysis	Applies the same reporting structure to the combined-block analyses without serving as the main empirical basis of the thesis

One further clarification is useful here. The dataset-audit stage is not merely descriptive: it also includes a grouped-CV classical baseline in which each fold uses training-only standardization, training-fold PCA, and `saga`-based logistic regression. This procedure can run in either the original three-class mode or a left-versus-right mode. In the present comparison set, however, the paired no-tail and with-tail datasets are absent, so this baseline cannot yet be reported with completed results.

E.4 Supplementary visual coverage

Appendix D gathers the supplementary diagnostic figures. These grids make it possible to verify that each analysis includes its own compact validation-versus-test plots, train/validation learning curves, learning-rate schedule, and final confusion matrix.

For completeness, the figures were also reviewed as contact sheets during manuscript preparation so that the appendix captions and the surrounding discussion could be checked against the underlying visuals.

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